Microsoft

WorkshopPlus: Azure AI Platform and Services

AI Language Services

Conditions and terms of use

Learning Units covered in this Module

- Introduction to AI Language Services
- Natural Language Processing: Extract and classify text, understand conversations, AI Translator, Content Understanding/AI Foundry
- Choosing the right AI Service
- Responsible Use
- Containerization, Integration, Real World applications and AI Agents
- Lab: AI Language

Introduction to AI Language Services

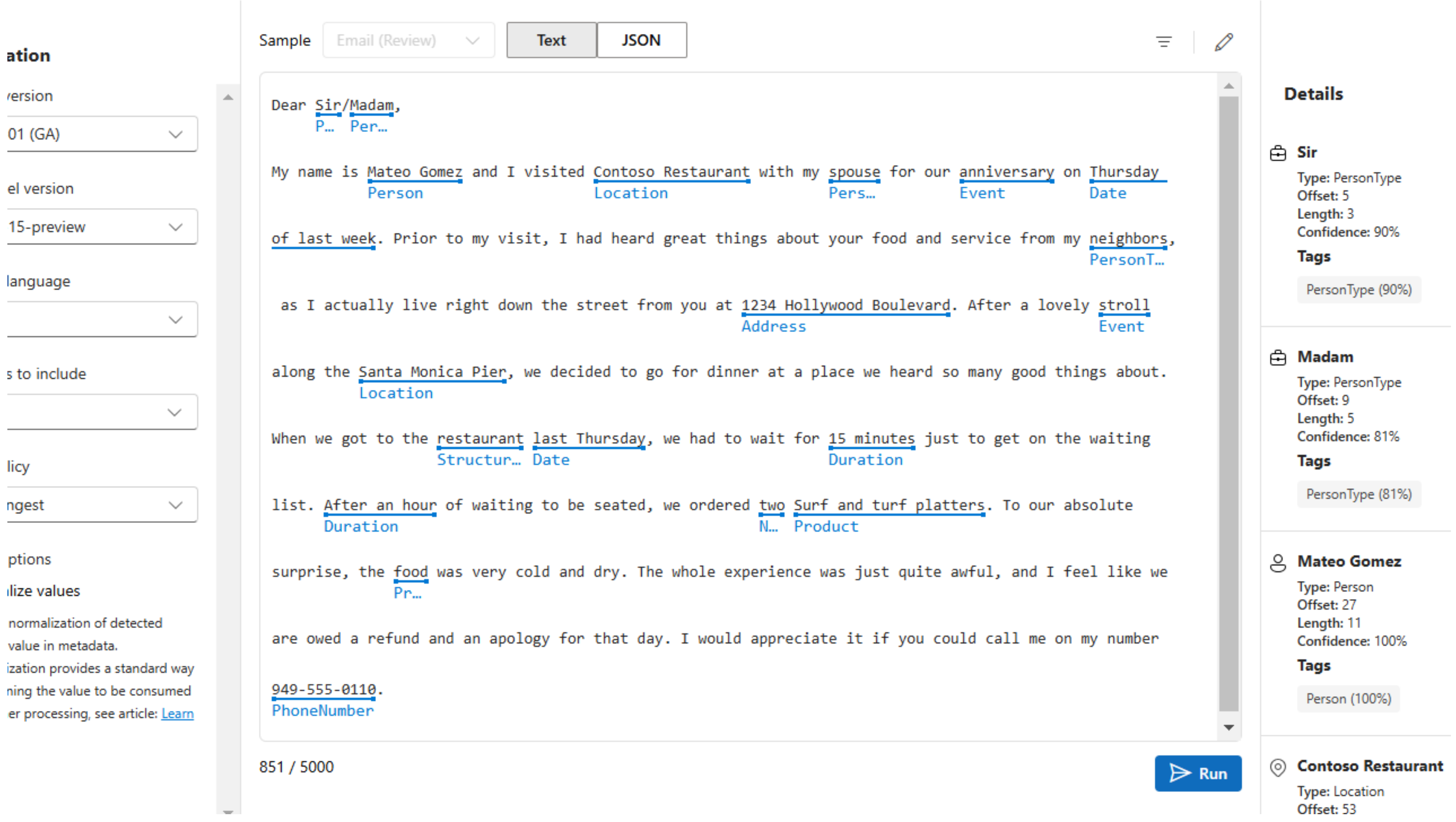
Objectives

After completing this Learning, you will be able to:

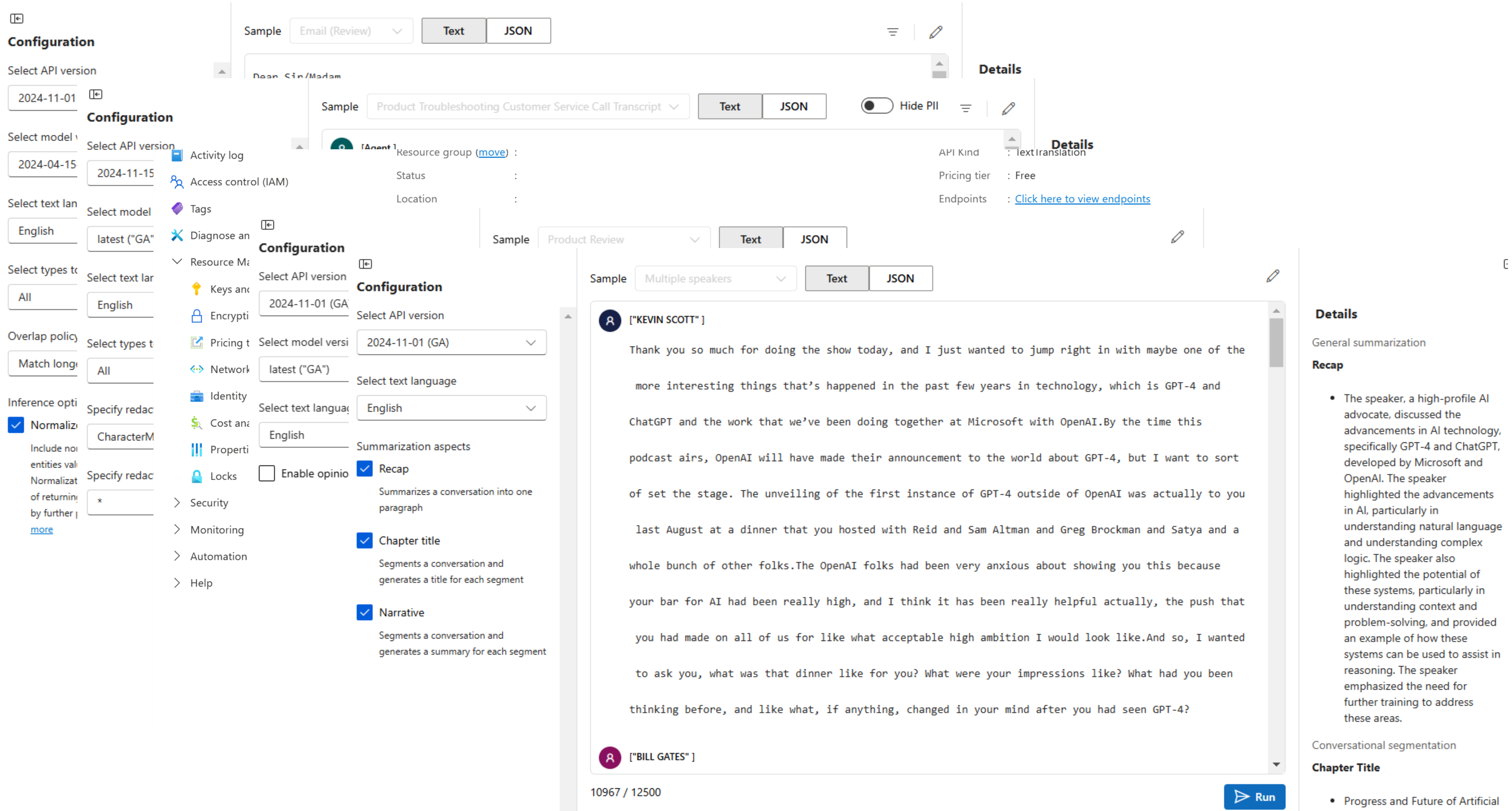
1	**Understand....** ✓ The different Azure AI Language Services ✓ How these services work
2	**Identify....** ✓ Use cases for each service

What is Natural Language Processing (NLP)?

Azure AI Language is a cloud-based service that provides **Natural Language Processing** (NLP) features for **understanding** and **analyzing** text.



AI language examples



Microsoft is powered by Azure AI

Applications

Microsoft 365 | Microsoft Dynamics 365 | Copilot | Partner Solutions

Application Platform
AI Builder / Low Code

Power BI | Power Apps | Power Automate | Copilot Studio Copilot Agents

Scenario-Based Services

Bot Service | AI Search | Document Intelligence | Video Indexer | Metrics Advisor | Immersive Reader

Customizable AI Models

Vision | Speech | Language | Decision | Azure OpenAI Service + Industry Models

ML Platform

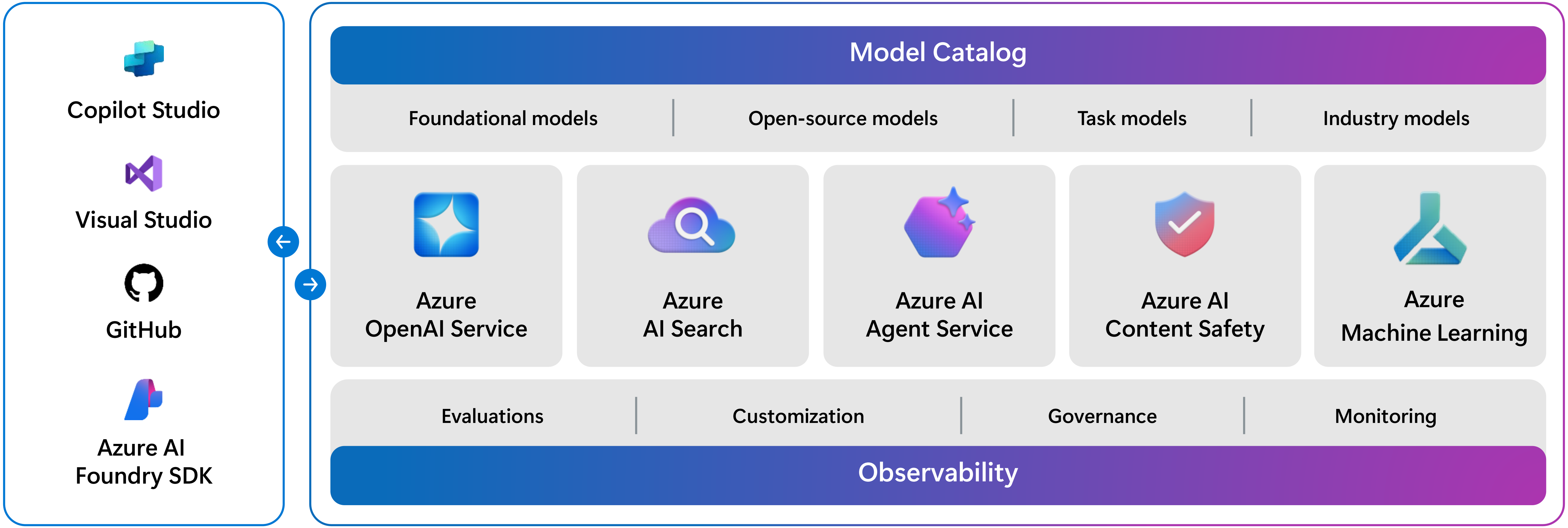
Azure Machine Learning | Microsoft Fabric

Microsoft products using Azure AI Language

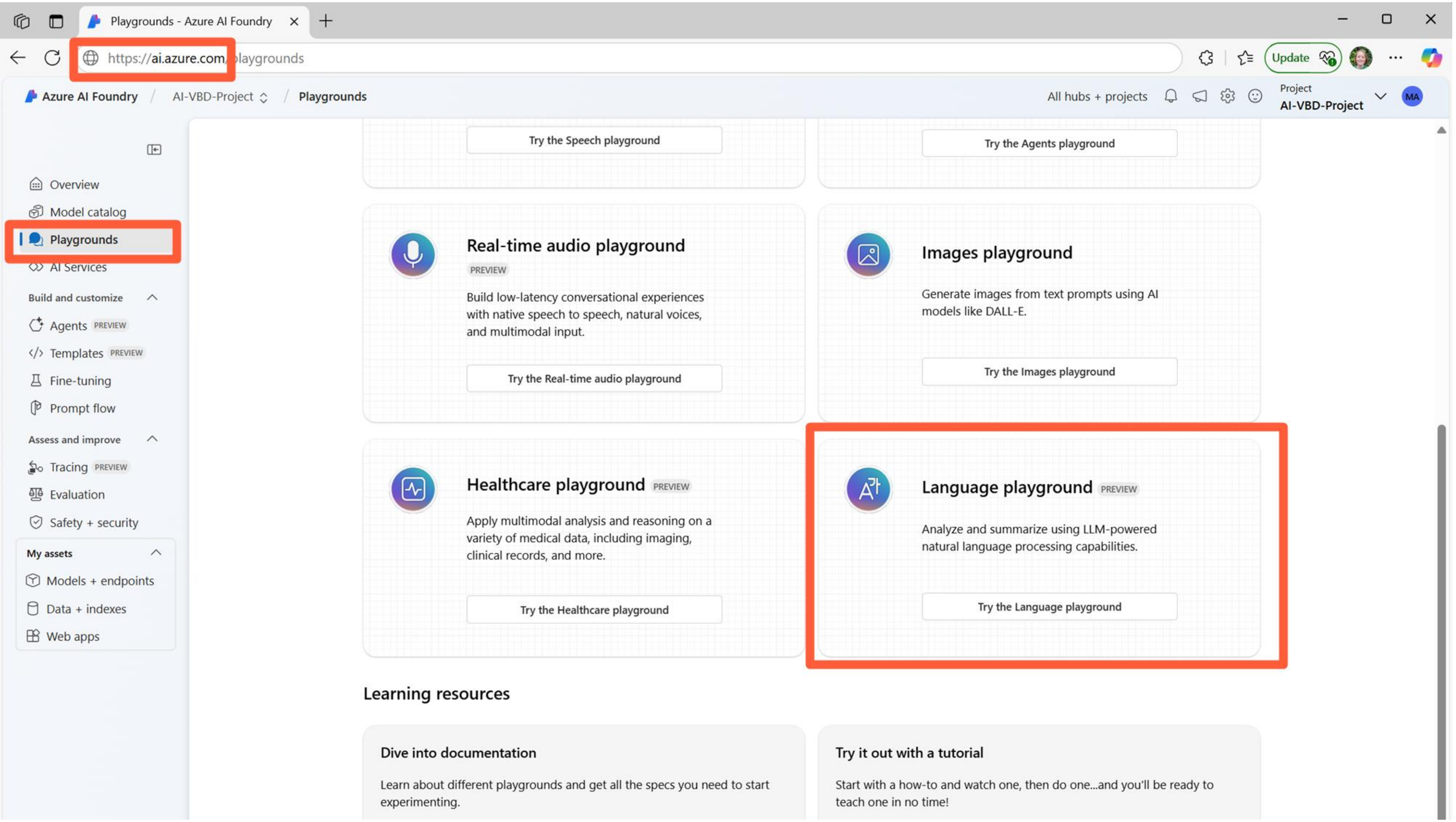
Bing	Office	SharePoint	Teams	Purview
Viva Engage	Dynamics 365	Designer	Visual Studio	LinkedIn

Coming Soon: AI Interpreter Agent

Azure AI Foundry



AI Foundry


Playgrounds - Azure AI Foundry
https://ai.azure.com/playgrounds
Update
Azure AI Foundry / AI-VBD-Project / Playgrounds
All hubs + projects
Project
AI-VBD-Project
Overview
Model catalog
Playgrounds
AI Services
Build and customize
Agents PREVIEW
Templates PREVIEW
Fine-tuning
Prompt flow
Assess and improve
Tracing PREVIEW
Evaluation
Safety + security
My assets
Models + endpoints
Data + indexes
Web apps
Try the Speech playground
Try the Agents playground
Real-time audio playground
PREVIEW
Build low-latency conversational experiences with native speech to speech, natural voices, and multimodal input.
Try the Real-time audio playground
Images playground
Generate images from text prompts using AI models like DALL-E.
Try the Images playground
Healthcare playground PREVIEW
Apply multimodal analysis and reasoning on a variety of medical data, including imaging, clinical records, and more.
Try the Healthcare playground
Language playground PREVIEW
Analyze and summarize using LLM-powered natural language processing capabilities.
Try the Language playground
Learning resources
Dive into documentation
Learn about different playgrounds and get all the specs you need to start experimenting.
Try it out with a tutorial
Start with a how-to and watch one, then do one...and you'll be ready to teach one in no time!

Natural Language Processing

NLP Objectives

After completing this Learning, you will be able to:

1	**Understand….** ✓ The principles of natural language processing ✓ The principles of text summarisation, translation, information extraction
2	**Identify….** ✓ Use cases for each service

Azure AI Language Services

Extract, Classify, Summarize

- Extract key phrases
- Find linked entities
- Named Entity Recognition (NER)
- Personally Identifiable Information (PII) detection
- Custom Named Entity Recognition (custom NER)
- Text analytics for health
- Document and Conversation summarization
- Analyze sentiment and mine text for opinions
- Detect Language
- Custom text classification

Translate

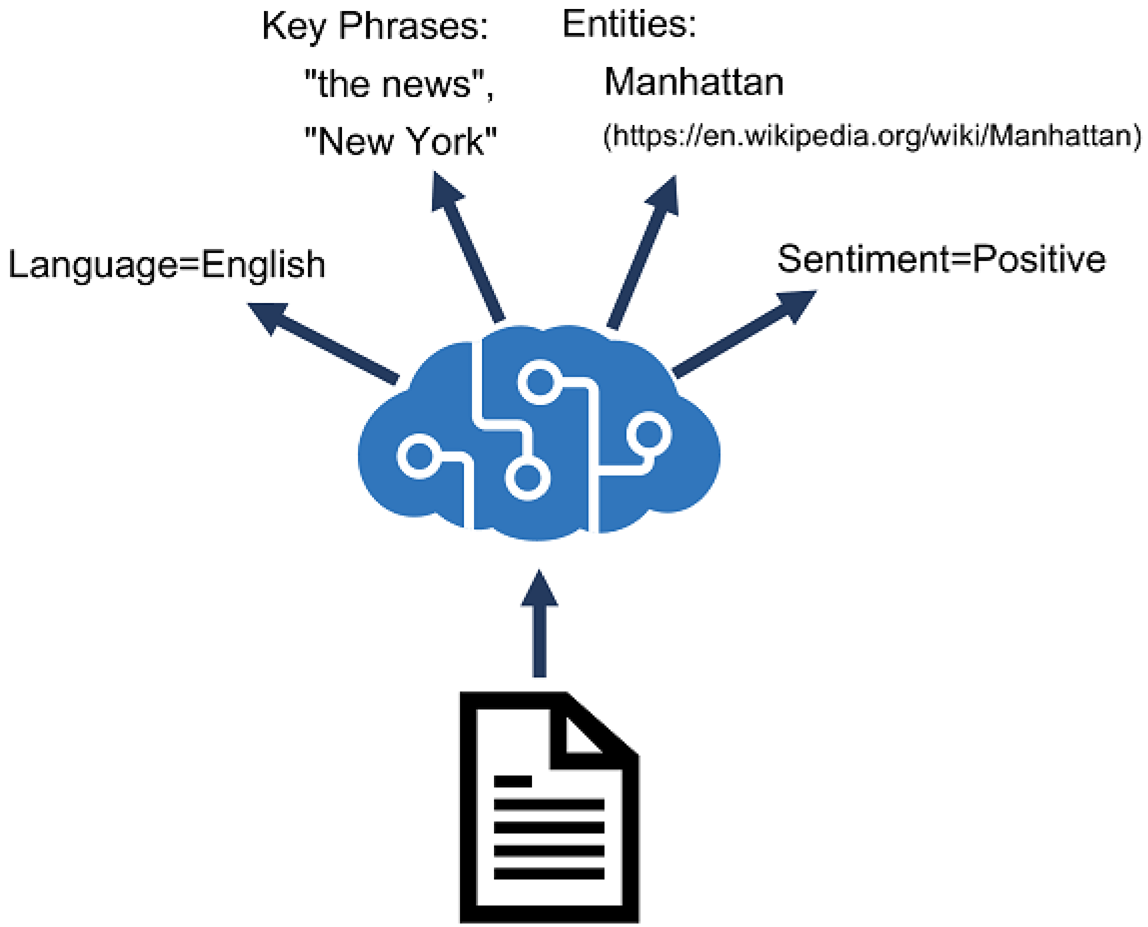
- Text Translation
- Document Translation
- Custom Translator

Language Understanding

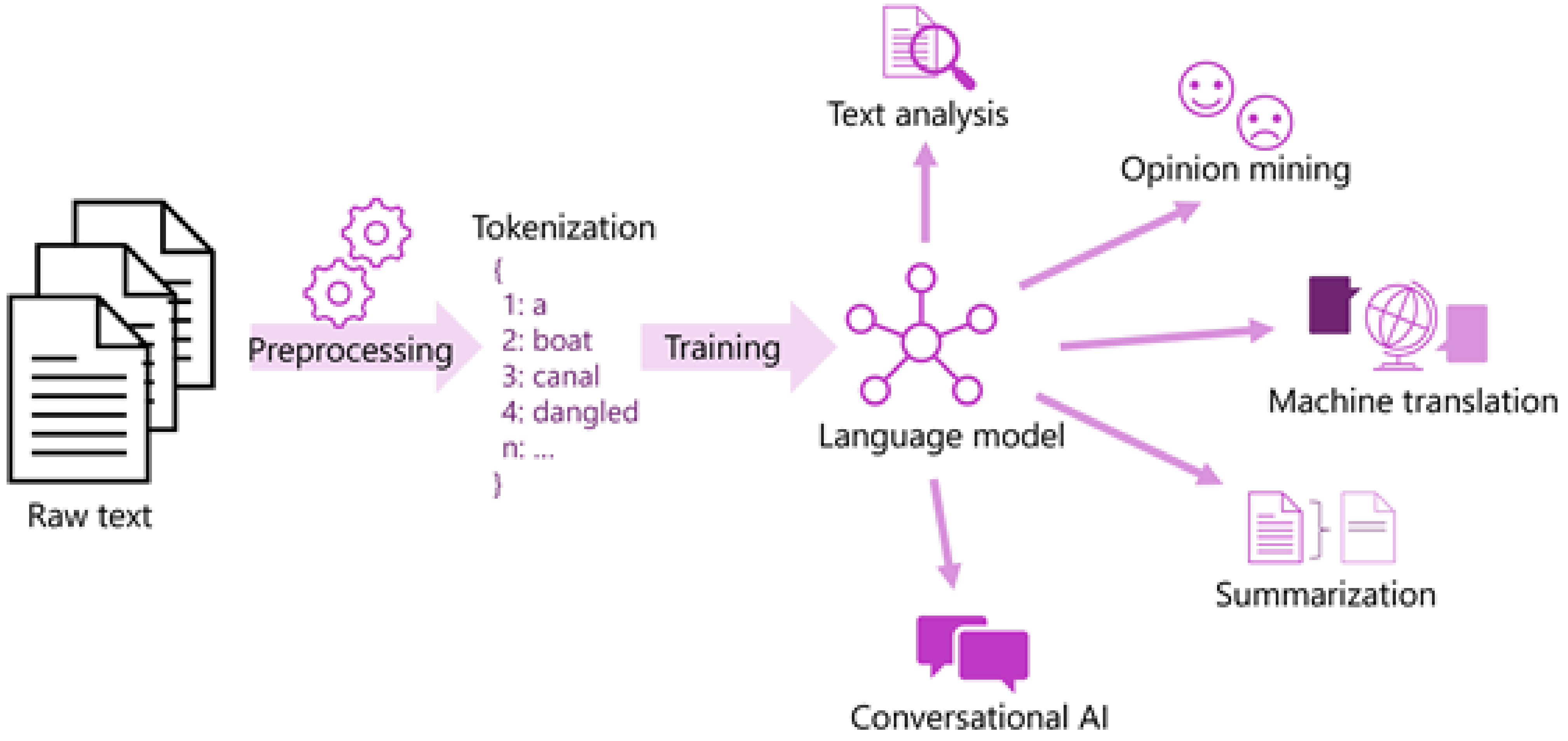
- Conversational Language Understanding
- Orchestration Workflow
- Question Answering

AI Language

Azure AI Language is a cloud-based service that provides Natural Language Processing (NLP) features for understanding and analyzing text.



Natural language processing overview



Tokenization

We choose to go to the moon.

we choose to go to the moon

Text normalisation

we choose to go to the moon

Tokenization

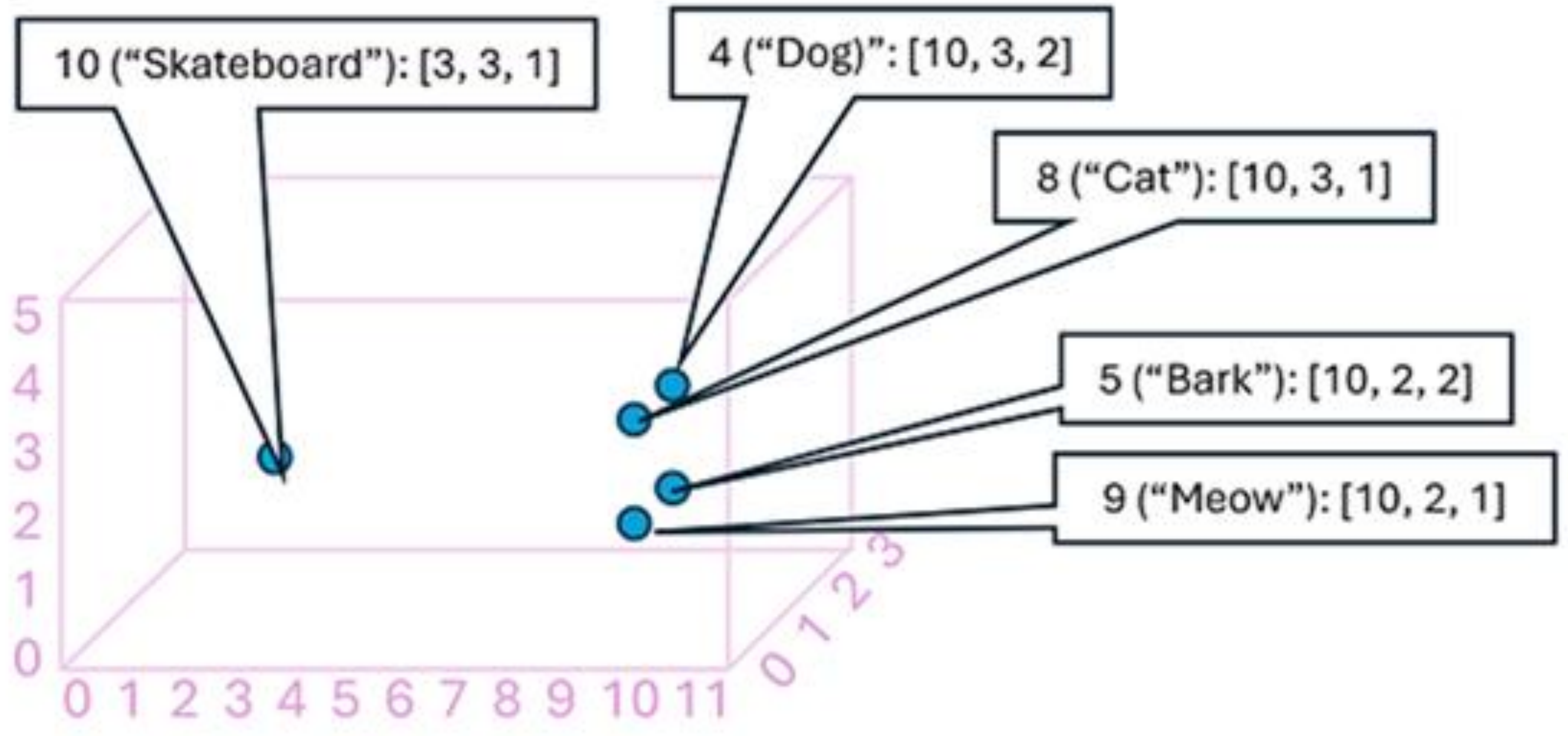
we choose ~~to~~ go ~~to the~~ moon

Stop word removal

we choose ~~to~~ go ~~to the~~ moon

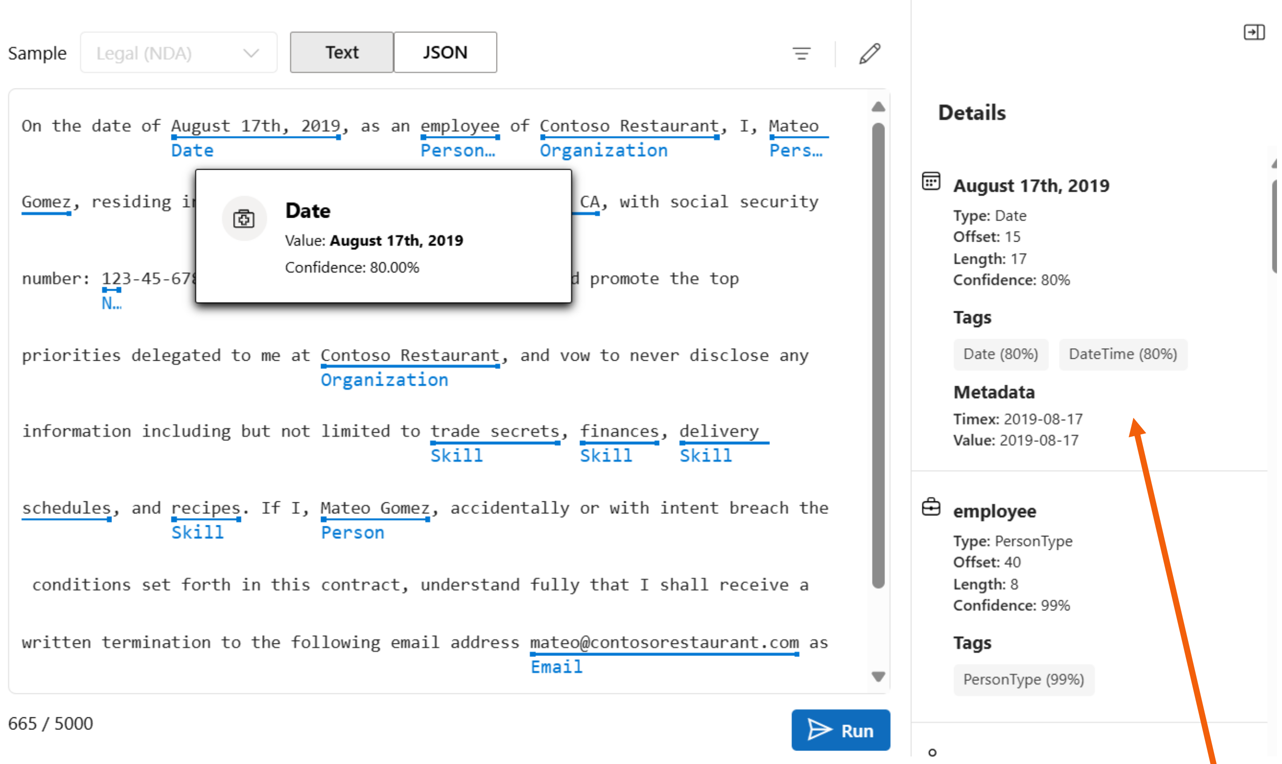
n-grams ***(bigram)***

Semantic Language Models



Named Entity Recognition (and custom)

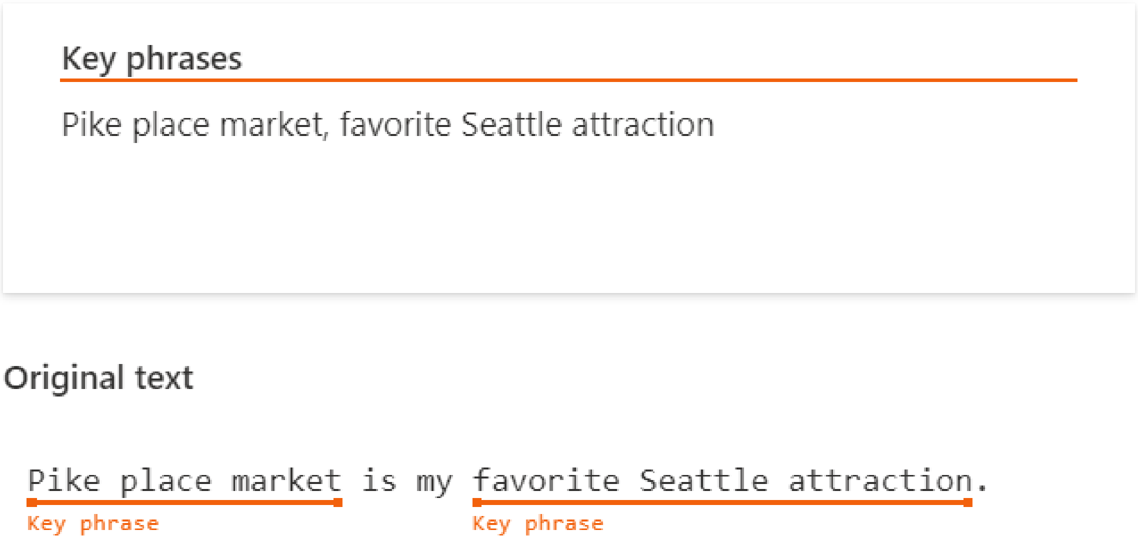
- News aggregation
- Analysing customer queries
- Research tool
- Document analysis
- Enhance search capabilities
- Automate business processes
- Customer analysis



Note confidence score and ISO formatted date

Key phrase extraction

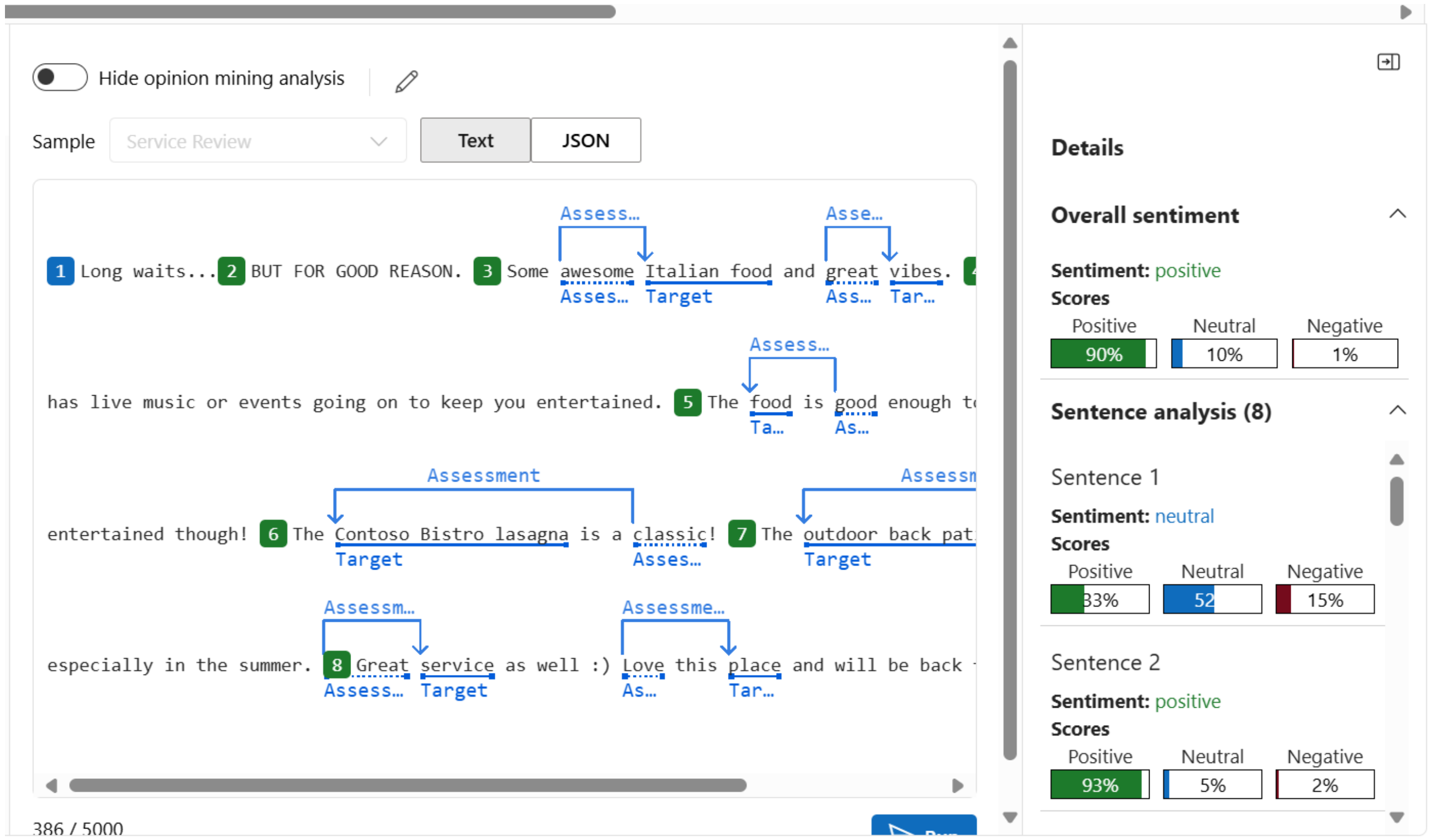
- Information Retrieval
- Text Summarisation
- Market research
- Customer support categorisation/prioritisation
- Content Recommendation



Sentiment Analysis and Opinion Mining

Example Applications

- Customer feedback analysis
- Social media analysis
- Product development
- Brand monitoring
- Survey analysis
- Customer service improvement
- Marketing campaign analysis



Entity Linking

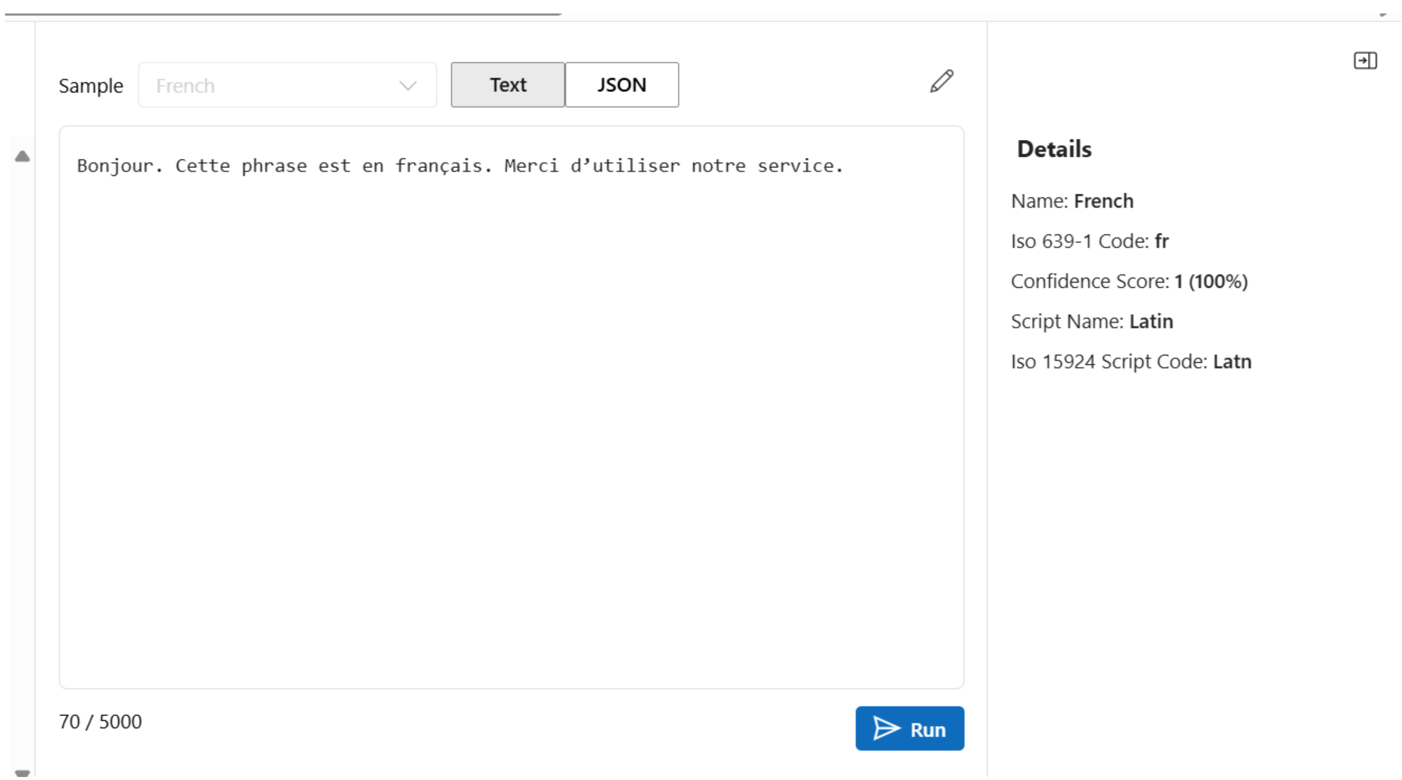
Example Applications

- Enhanced search engine understanding
- Improved content recommendation
- Information extraction
- Improving performance of chatbots and virtual assistants by linking entities to a knowledge base

Language Detection

Example Applications

- Customer service routing
- Content translation
- Security and compliance (eg for PII)
- Social media analysis
- Identify language of reviews
- Document management



SDK and REST API for the Language Service

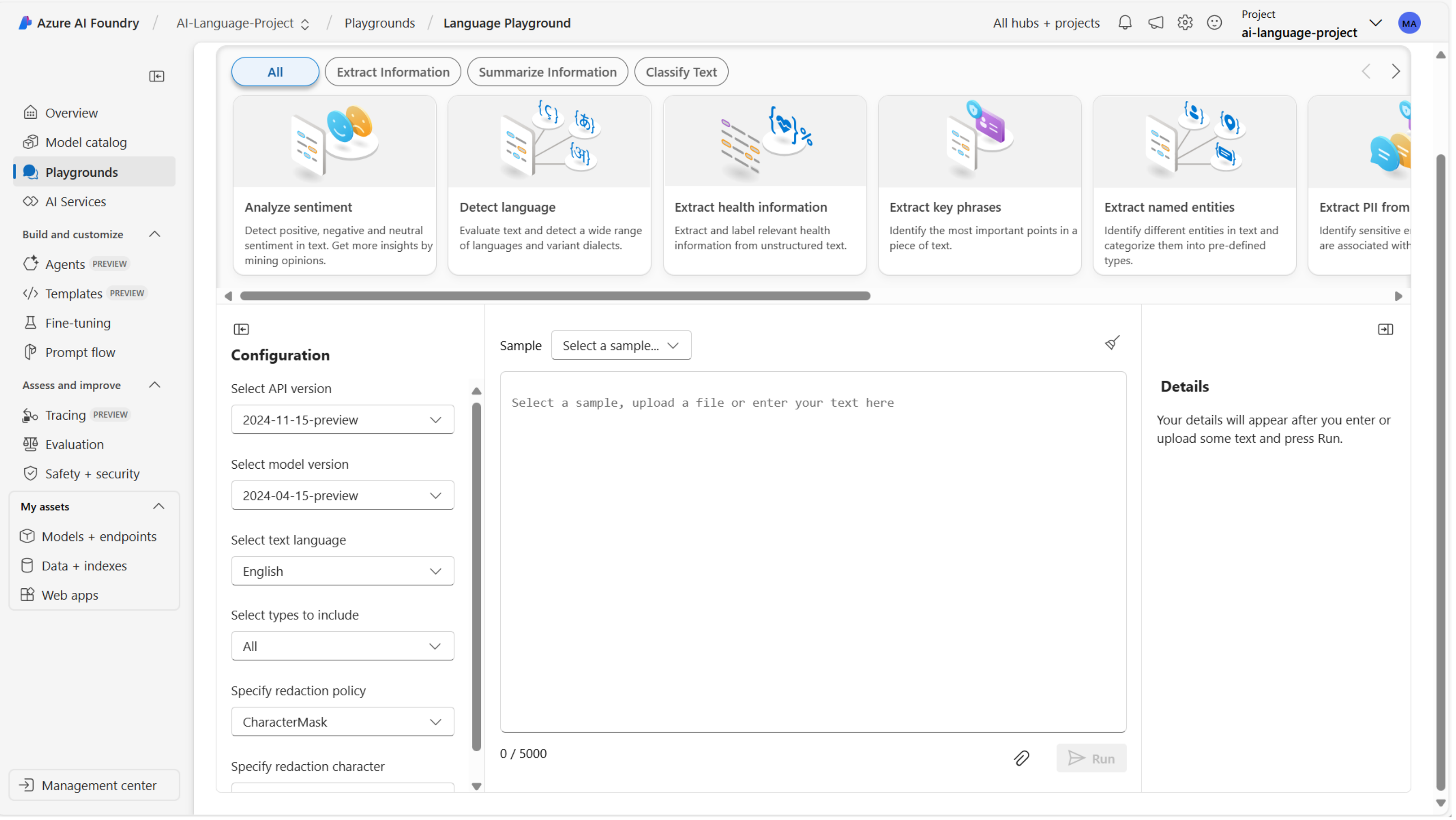
Client libraries available for

- .NET
- Java
- Javascript
- Python

https://learn.microsoft.com/en-us/azure/ai-services/language-service/concepts/developer-guide?tabs=language-studio

Demonstration

AI Language Extraction



Summarization

Summarization Objectives

After completing this Learning, you will be able to:

1	**Understand....** ✓ The principles of summarization ✓ The principles of extractive and abstractive summarization
2	**Identify....** ✓ Use cases for each service

Summarize

Example Applications

- Customer service summaries
- Meeting summaries
- Content creation
- Document management
- News aggregation
- Summarise educational content

Extractive and Abstractive Summarization

Abstractive text summarization now using Phi-3.5-mini!

Input

"At Microsoft, we are on a quest to advance AI beyond existing techniques, by taking a more holistic, human-centric approach to learning and understanding. As Chief Technology Officer of Azure AI services, I have been working with a team of amazing scientists and engineers to turn this quest into a reality. In my role, I enjoy a unique perspective in viewing the relationship among three attributes of human cognition: monolingual text (X), audio or visual sensory signals, (Y) and multilingual (Z). At the intersection of all three, there's magic—what we call XYZ-code as illustrated in Figure 1—a joint representation to create more powerful AI that can speak, hear, see, and understand humans better. We believe XYZ-code enables us to fulfill our long-term vision: cross-domain transfer learning, spanning modalities and languages. The goal is to have pretrained models that can jointly learn representations to support a broad range of downstream AI tasks, much in the way humans do today. Over the past five years, we achieve human performance on benchmarks in conversational speech recognition, machine translation, conversational question answering, machine reading comprehension, and image captioning. These five breakthroughs provided us with strong signals toward our more ambitious aspiration to produce a leap in AI capabilities, achieving multi-sensory and multilingual learning that is closer in line with how humans learn and understand. I believe the joint XYZ-code is a foundational component of this aspiration, if grounded with external knowledge sources in the downstream AI tasks."

Extractive

- "At Microsoft, we are on a quest to advance AI beyond existing techniques, by taking a more holistic, human-centric approach to learning and understanding."
- "We believe XYZ-code enables us to fulfill our long-term vision: cross-domain transfer learning, spanning modalities and languages."
- "The goal is to have pretrained models that can jointly learn representations to support a broad range of downstream AI tasks, much in the way humans do today."

Abstractive

"Microsoft is taking a more holistic, human-centric approach to learning and understanding. We believe XYZ-code enables us to fulfill our long-term vision: cross-domain transfer learning, spanning modalities and languages. Over the past five years, we achieved human performance on benchmarks in conversational speech recognition."

Conversation Summarization

Input

Agent: "*Hello, you're chatting with Rene. How may I help you?*"

Customer: "*Hi, I tried to set up wifi connection for Smart Brew 300 espresso machine, but it didn't work.*"

Agent: "*I'm sorry to hear that. Let's see what we can do to fix this issue. Could you push the wifi connection button, hold for 3 seconds, then let me know if the power light is slowly blinking?*"

Customer: "*Yes, I pushed the wifi connection button, and now the power light is slowly blinking.*"

Agent: "*Great. Thank you! Now, please check in your Contoso Coffee app. Does it prompt to ask you to connect with the machine?*"

Customer: "*No. Nothing happened.*"

Agent: "*I see. Thanks. Let's try if a factory reset can solve the issue. Could you please press and hold the center button for 5 seconds to start the factory reset.*"

Customer: "*I've tried the factory reset and followed the above steps again, but it still didn't work.*"

- **Agent**: "*I'm very sorry to hear that. Let me see if there's another way to fix the issue. Please hold on for a minute.*"

Output

Example summary	Remark	Conversation aspect
Customer is unable to set up wifi connection for Smart Brew 300 espresso machine	a customer issue in a customer-and-agent conversation	issue
The agent suggested several troubleshooting steps, including checking the wifi connection, checking the Contoso Coffee app, and performing a factory reset. However, none of these steps resolved the issue. The agent then put the customer on hold to look for another solution.	solutions tried in a customer-and-agent conversation	resolution
The customer contacted the agent for assistance with setting up a wifi connection for their Smart Brew 300 espresso machine. The agent guided the customer through several troubleshooting steps, including a wifi connection check, checking the power light, and a factory reset. Despite following these steps, the issue persisted. The agent then decided to explore other potential solutions	Summarizes a conversation into one paragraph	recap
Troubleshooting SmartBrew 300 Espresso Machine	Segments a conversation and generates a title for each segment; usually cowork with narrative aspect	chapterTitle
The customer is having trouble setting up a wifi connection for their Smart Brew 300 espresso machine. The agent suggests several solutions, including a factory reset, but the issue persists.	Segments a conversation and generates a summary for each segment, usually cowork with chapterTitle aspect	narrative

Document Summarization - *Preview*

File type	File extension	Description
Text	.txt	An unformatted text document.
Adobe PDF	.pdf	A portable document file formatted document.
Microsoft Word	.docx	A Microsoft Word document file.

Why does Azure AI Language's Summarization Service matter?

Premium Quality at Affordable Price: The combination of Generative Language AI models (Azure Open AI service) and task-optimized encoder models offers text summarization solutions with higher quality, cheaper costs, and lower latency.

1. Azure AI Language's Summarization - Key Customer Scenarios

1.1 Summarization - Conversation

Contact Center Call & Meeting Analytics

Microsoft Teams

comcast

1.2 Summarization - Document

Document Search & Document Discovery

ARTHUR D LITTLE

1. Azure AI Language's Summarization - Key Highlights

10x Cost Savings: Azure AI Language Summarization in English costs ~0.008/1k tokens vs. $0.06/1k tokens for AOAI/GPT4-32k

Leading Quality with fine-tuned LLM for both document and conversation summarization with built-in hallucination detection.

More Capabilities: Extractive (encoder model) + **Abstractive** summarization enables more use cases than decoder-only Gen AI

No Prompt Engineering needed because of out-of-the-box summarization styles and aspects with native document support

Disconnected & Connected Container offerings with both GPU & CPU options with distilled models for optimal compute usage

Summarization Demonstration

Summarization in AI Language

PII Service

PII Objectives

After completing this Learning, you will be able to:

1	**Understand….** ✓ The principles of Personal Identifying Information (PII) detection ✓ The principles of PII Redaction/Removal
2	**Identify….** ✓ Use cases for each service

PII Detection

URL

Entity value:
www.contososteakhouse.com
Confidence: 80.00%

PhoneNumber

Entity value: 312-555-0176
Confidence: 80.00%

Email

Entity value:
order@contososteakhouse.com
Confidence: 80.00%

Organization

Entity value:
contososteakhouse
Confidence: 46.00%

Original text

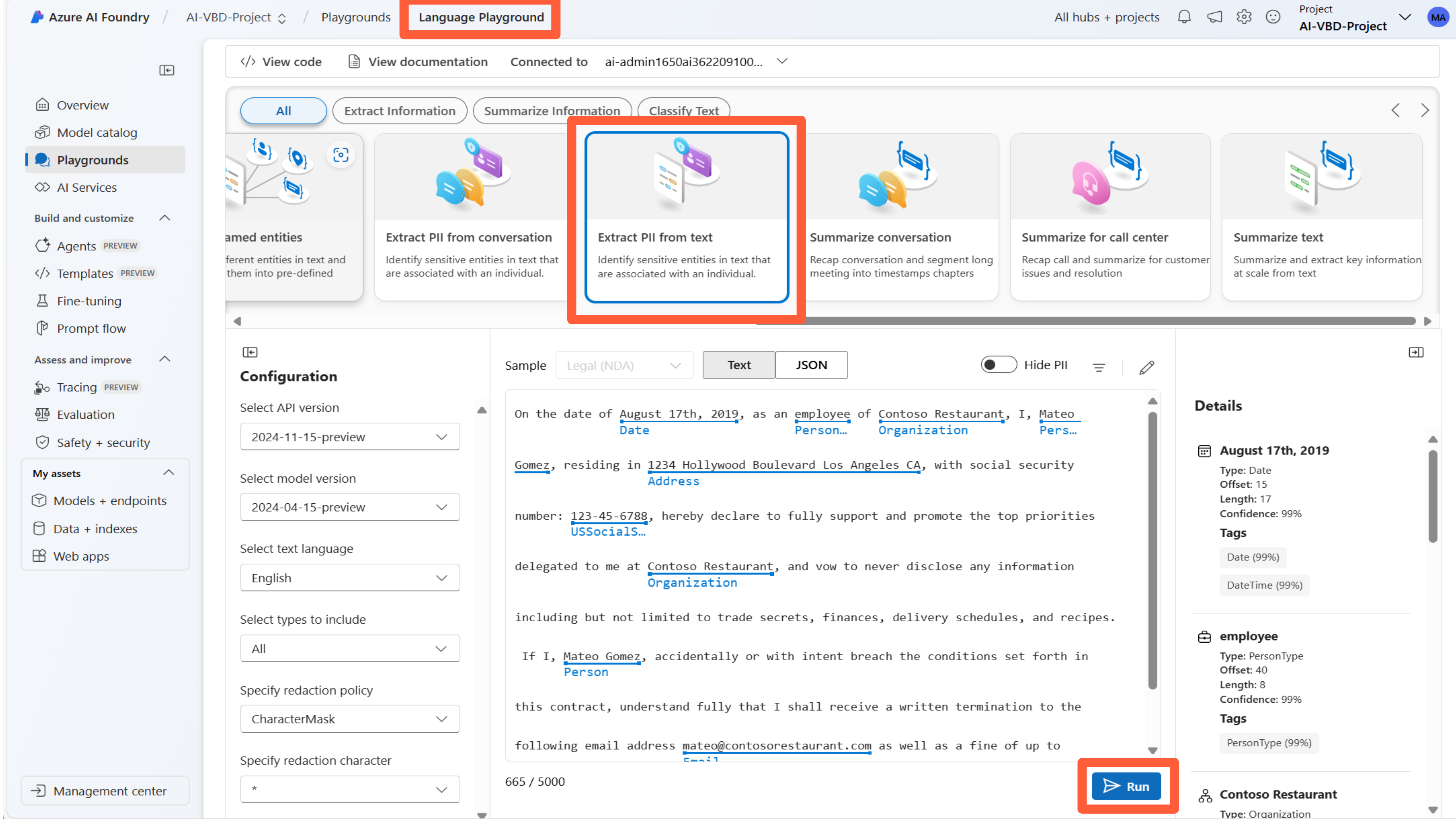
You can even pre-order from their online menu at www.contososteakhouse.com, call 312-555-0176 or send email to order@contososteakhouse.com!

URL

PhoneNumber

Organization

Email


Azure AI Foundry
AI-VBD-Project
Playgrounds
Language Playground
All hubs + projects
Project
AI-VBD-Project
MA
View code
View documentation
Connected to
ai-admin1650ai362209100...
Overview
Model catalog
Playgrounds
AI Services
Build and customize
Agents PREVIEW
Templates PREVIEW
Fine-tuning
Prompt flow
Assess and improve
Tracing PREVIEW
Evaluation
Safety + security
My assets
Models + endpoints
Data + indexes
Web apps
Management center
All
Extract Information
Summarize Information
Classify Text
amed entities
ferent entities in text and them into pre-defined
Extract PII from conversation
Identify sensitive entities in text that are associated with an individual.
Extract PII from text
Identify sensitive entities in text that are associated with an individual.
Summarize conversation
Recap conversation and segment long meeting into timestamps chapters
Summarize for call center
Recap call and summarize for customer issues and resolution
Summarize text
Summarize and extract key information at scale from text
Configuration
Select API version
2024-11-15-preview
Select model version
2024-04-15-preview
Select text language
English
Select types to include
All
Specify redaction policy
CharacterMask
Specify redaction character
*
Sample
Legal (NDA)
Text
JSON
Hide PII
On the date of August 17th, 2019, as an employee of Contoso Restaurant, I, Mateo
Date
Person...
Organization
Pers...
Gomez, residing in 1234 Hollywood Boulevard Los Angeles CA, with social security
Address
number: 123-45-6788, hereby declare to fully support and promote the top priorities
USSocialS...
delegated to me at Contoso Restaurant, and vow to never disclose any information
Organization
including but not limited to trade secrets, finances, delivery schedules, and recipes.
If I, Mateo Gomez, accidentally or with intent breach the conditions set forth in
Person
this contract, understand fully that I shall receive a written termination to the
following email address mateo@contosorestaurant.com as well as a fine of up to
665 / 5000
Run
Details
August 17th, 2019
Type: Date
Offset: 15
Length: 17
Confidence: 99%
Tags
Date (99%)
DateTime (99%)
employee
Type: PersonType
Offset: 40
Length: 8
Confidence: 99%
Tags
PersonType (99%)
Contoso Restaurant
Type: Organization

Data Anonymization

Challenge
Large volume of documents, forms and user queries with sensitive info poses high risks of data leaks.

Solution
Build an automated process to extract and anonymize sensitive information in documents, forms or user queries. Manage the extracted sensitive info in a separate, more secured environment.

Azure AI Services
- Personal Identifiable Information (PII) Detection with built-in native doc processing support
- Document Intelligence service

Impact
Data is protected more securely and efficiently.

Personally Identifiable Information (PII) Detection

- Identifies, categorizes, and redacts sensitive info in unstructured text
- Supports a wide range of PII info, including region-specific PII, PHI and PCI
- Supports conversational scenario with a specialized model (aka. Conversation PII)
- 79 languages supported

Typical use cases:

- Mask sensitive info to avoid the leak, inappropriate handling or data bias
- Auto label and classify docs based on the sensitivity

Sample text:	*Recognized entity categories:*
The food is great, and the place was impeccably clean. You can order from their online menu at www.contosohouse.com, call 312-555-0176 or send email to order@contosohouse.com!	*URL* *PhoneNumber* *Email*

Redacted text:

*The food is great, and the place was impeccably clean. You can order from their online menu at www.contosohouse.com, call ************ or send email to **********************!",*

PII Detection – *Supported Categories*

Basic PII Info
Person
Person Type
Phone Number
Organization – *incl. types:*
• Medical
• Stock Exchange
• Sports
Address
Email
URL
IP Address
Date Time – *incl. type:*
• Date
Quantity – *incl. type:*
• Age

Government & Country/Region-Specific
U.S. Social Security Number (SSN)
U.S. Driver's License Number
U.S. Passport Number
U.S. Individual Taxpayer Identification Number (ITIN)
U.S. Drug Enforcement Agency (DEA) Number
U.S. Bank Account Number
+ more than 90 PII entities across 31 other countries/regions

Financial Account
ABA Routing Number
SWIFT Code
Credit Card Number
IBAN Code

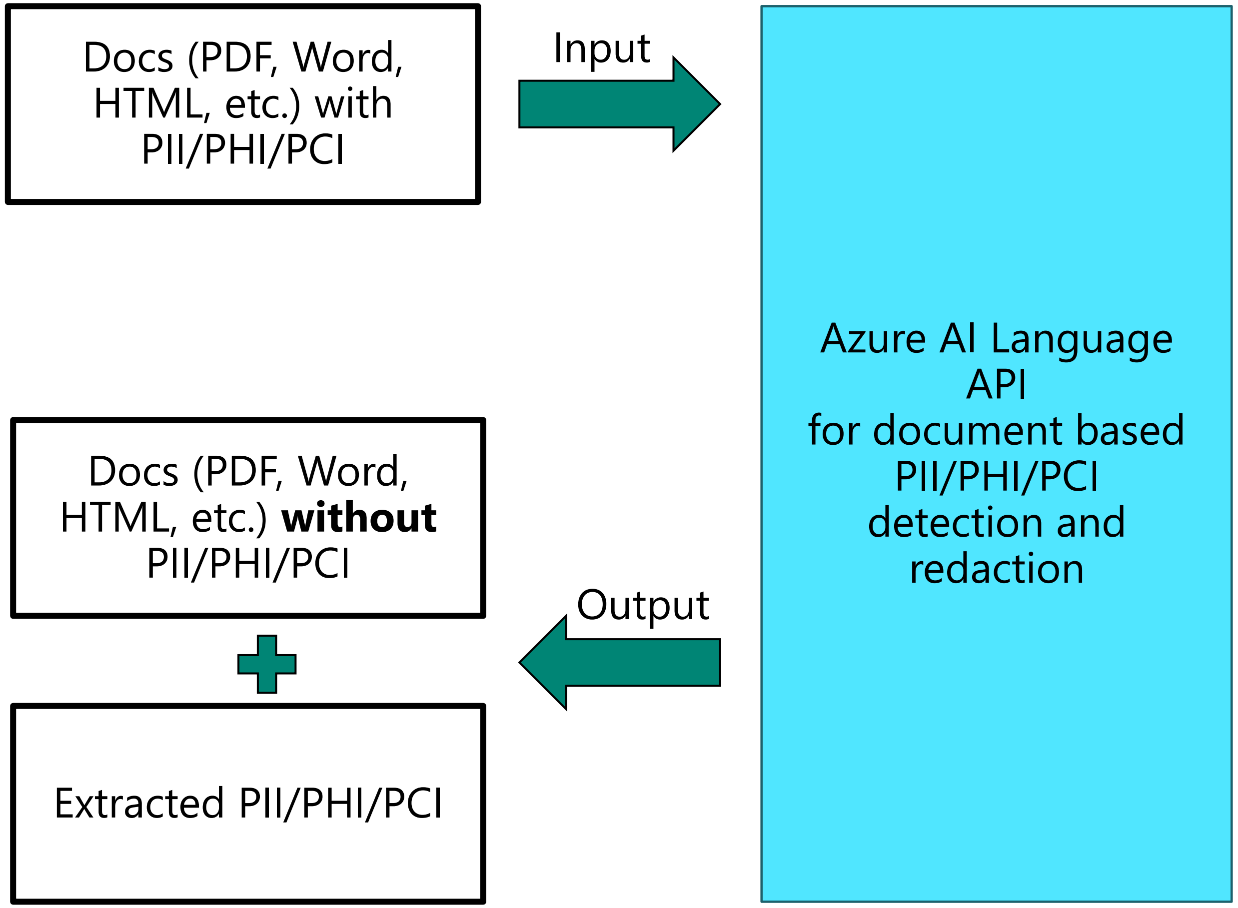
Azure Information
Azure DocumentDB Auth Key
Azure IAAS DB Connection String
Azure SQL Connection String
Azure IoT Connection String
Azure Publish Setting Password
Azure Redis Cache Connection String
Azure SAS
Azure Service Bus Connection String
Azure Storage Account Key
SQL Server Connection String

PII Redaction on Native Document Formats (preview)

- Identifies and redacts sensitive PII info directly in native documents –

  No more doc pre-processing and post-processing needed!

- PII entities and values extracted from documents are included in the API response payload, which can be stored separated and used for later



Why does Azure AI Language's PII Redaction matter?

Text-based PII unblocks adoption of Gen AI use cases by addressing the increased demand for PII. In fact, OpenAI's ChatGPT leverages Azure AI Language's PII service to preserve data privacy.

2. Azure AI Language' PII Service – Key Customer Scenarios

PII Redaction

Data Anonymization

OpenAI

DEPARTMENT OF THE NAVY · UNITED STATES OF AMERICA

2. Azure AI Language's PII Redaction – Key Highlights

2.1 Text-based PII Redaction

2.2 Native Document-based PII Redaction

2.2 Speech-based PII Redaction

TM

Differentiates Azure's Generative AI offerings from those of competitors (Google, AWS) by offering one solution for **Text-based PII, Document-based PII, and Conversational PII**

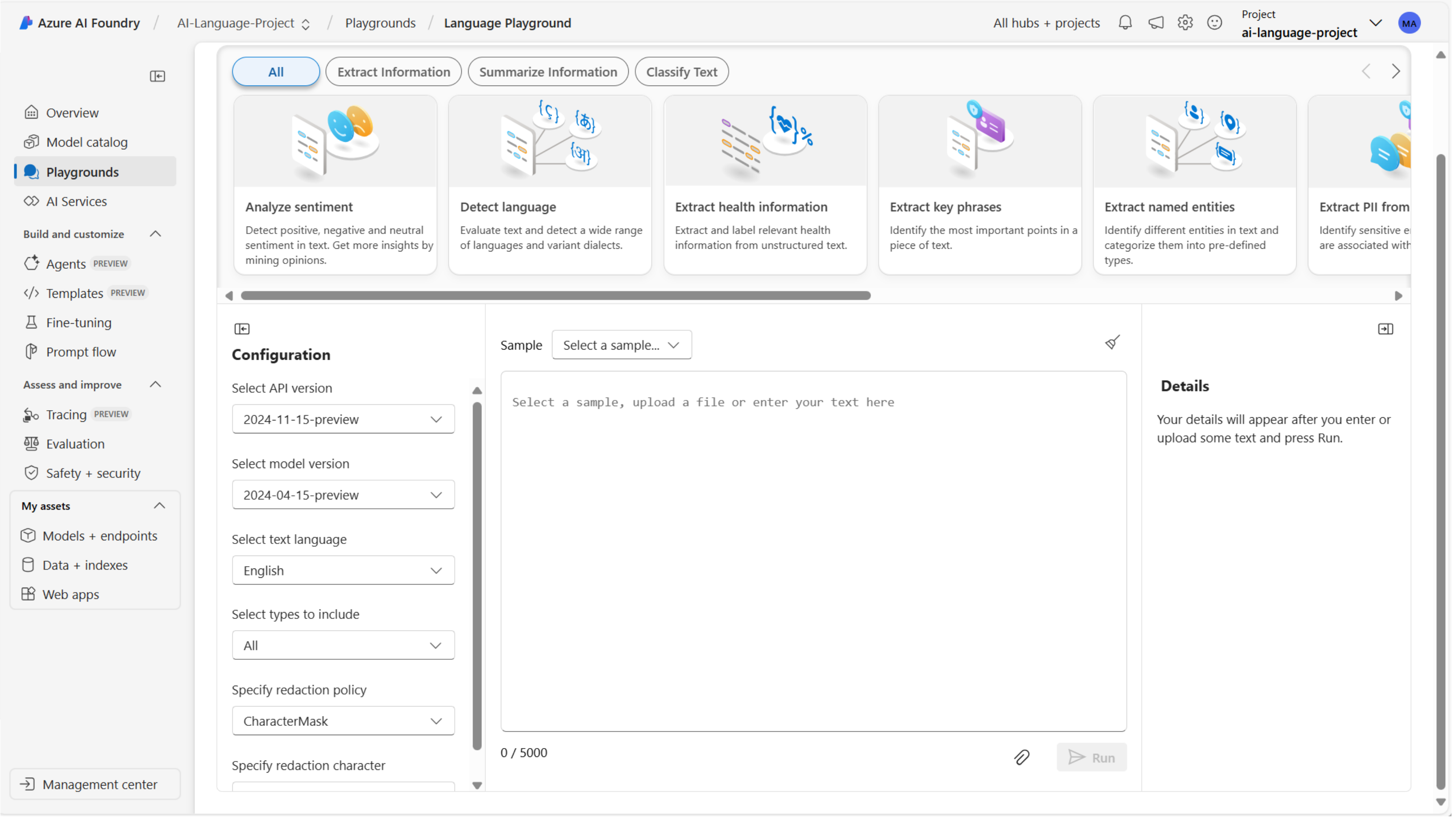
Document-based PII reduces the complexity to implement a PII detection and redaction solution by eliminating the need for document pre- or post-processing with **Single API call**

Conversational PII optimized for speech transcription use cases offers leading redaction quality to ensure sensitive information is redacted from both audio and speech transcript.

Ensures privacy and regulatory compliance for customers in industries such as insurance, finance, healthcare, and public sector to identify, detect & redact PII in large volume of data

PII Demonstration

PII Detection and Removal



Text Analytics for Health

Text Analytics for Health Objectives

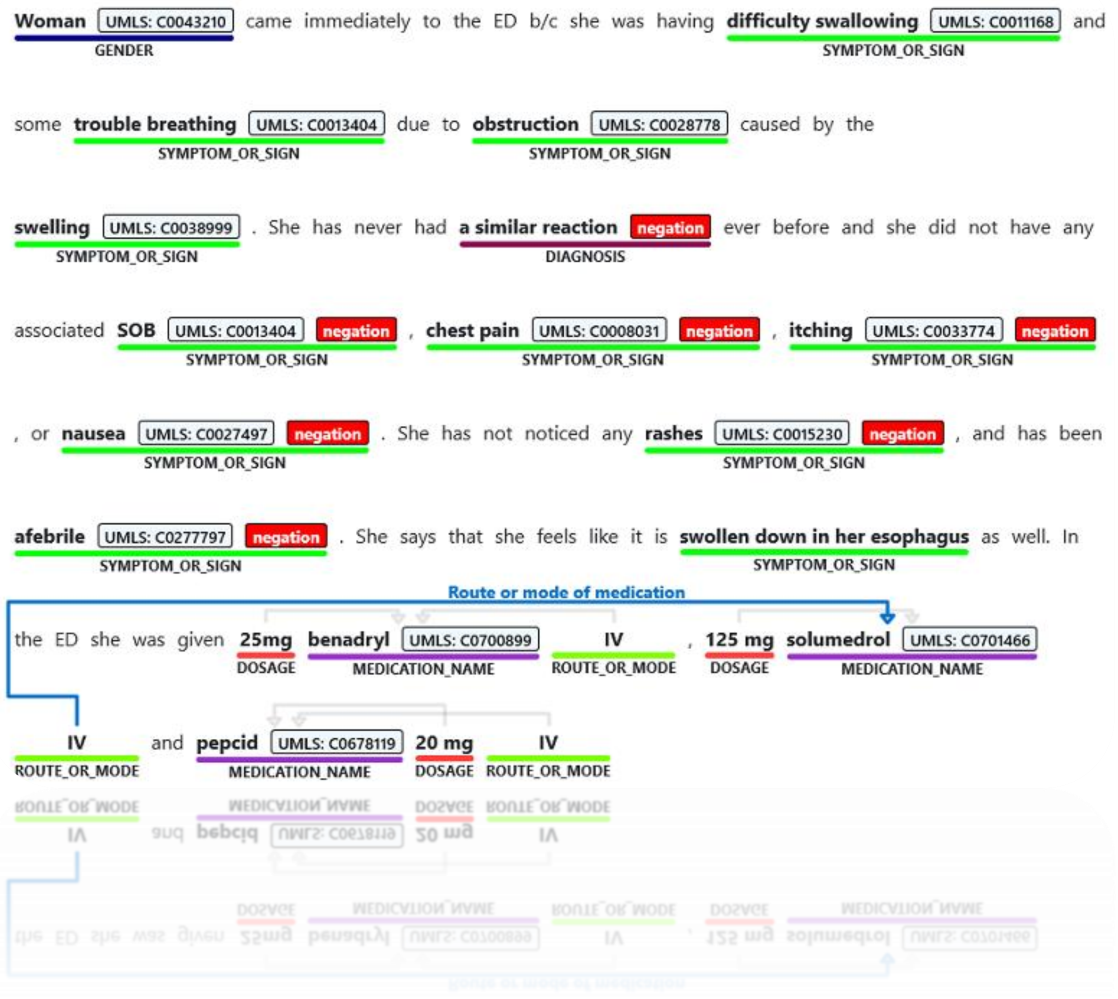
After completing this Learning, you will be able to:

1	**Understand....** ✓ The principles of the Text Analytics for Health entity recognition service ✓ The principles of relation and assertion in Text Analytics for Health
2	**Identify....** ✓ Use cases for this service

Text Analytics for health overview

Extracts and labels relevant medical info from unstructured texts such as doctor's notes, discharge summaries, clinical documents, and electronic health records

- **Entity recognition**
  (30+ healthcare related entity types)
- **Entity linking**
  (to UMLS Metathesaurus Knowledge Source)
- **Relation extraction** (35 relation types)
- **Assertion detection**
  (Certainty, Conditional, Association, Temporal)
- 7 languages supported (6 languages in preview)

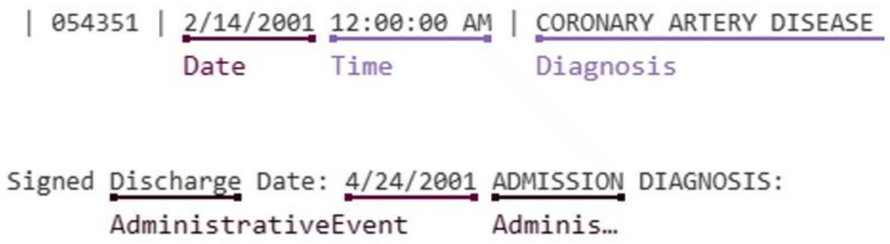


Available in containers

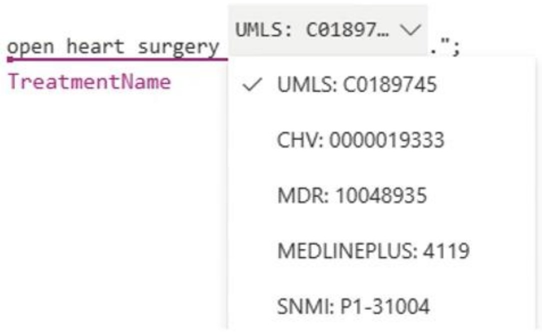
Demo: Text Analytics for Health

1 **Named Entity Recognition**:

Key terms highlighted & labelled

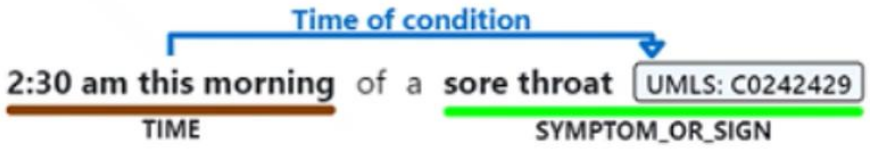


2 **Entity Linking**: Terms linked to respective UMLS codes



3 **Relation Extraction**:

Related terms associated together

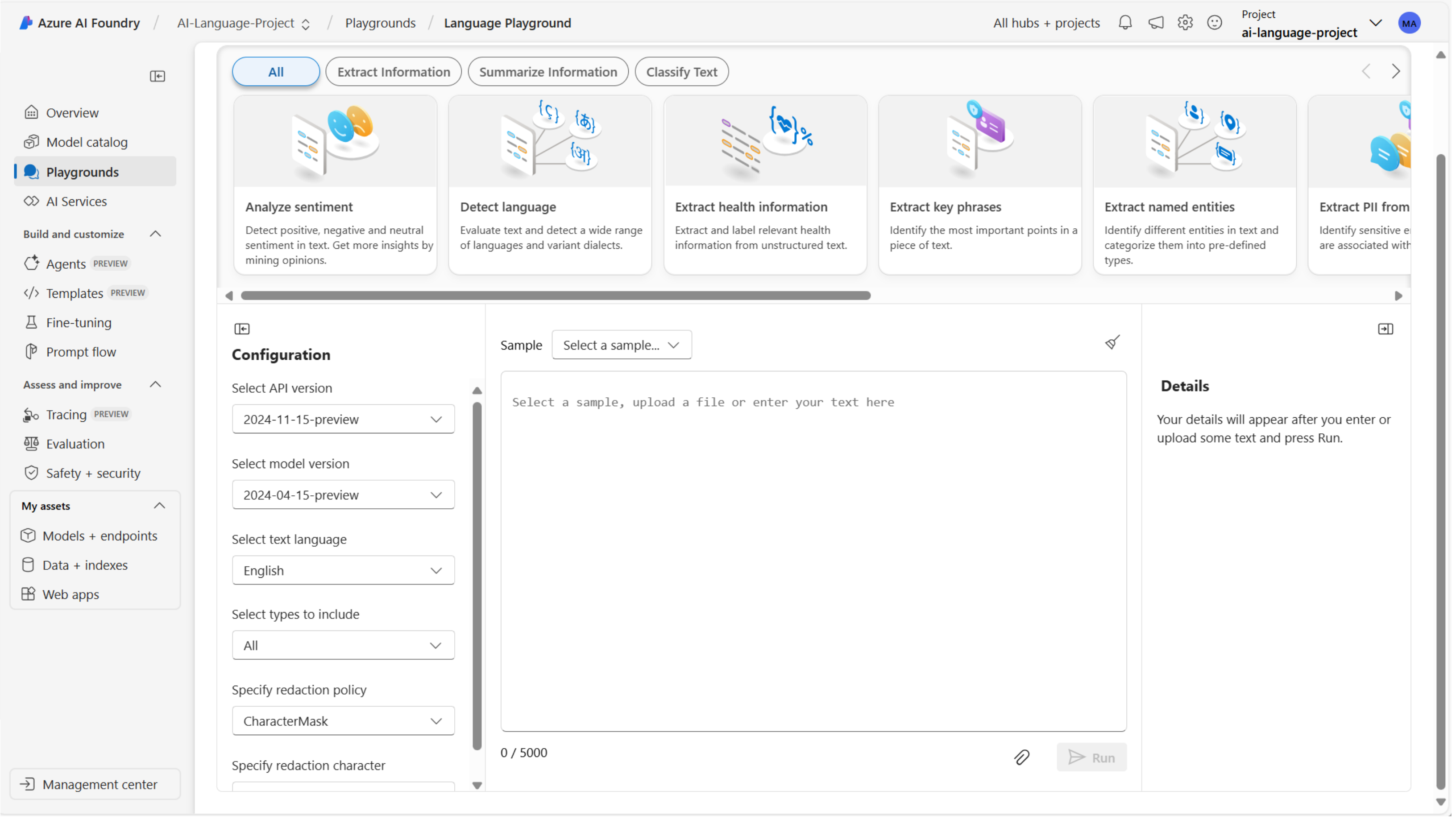


4 **Assertion Detection**:

Terms also labelled with contextual modifiers (i.e. certainty, conditionality, association)

Text Analytics for Health Demonstration

Text Analytics for Health



Language Understanding

Language Understanding Objectives

After completing this Learning, you will be able to:

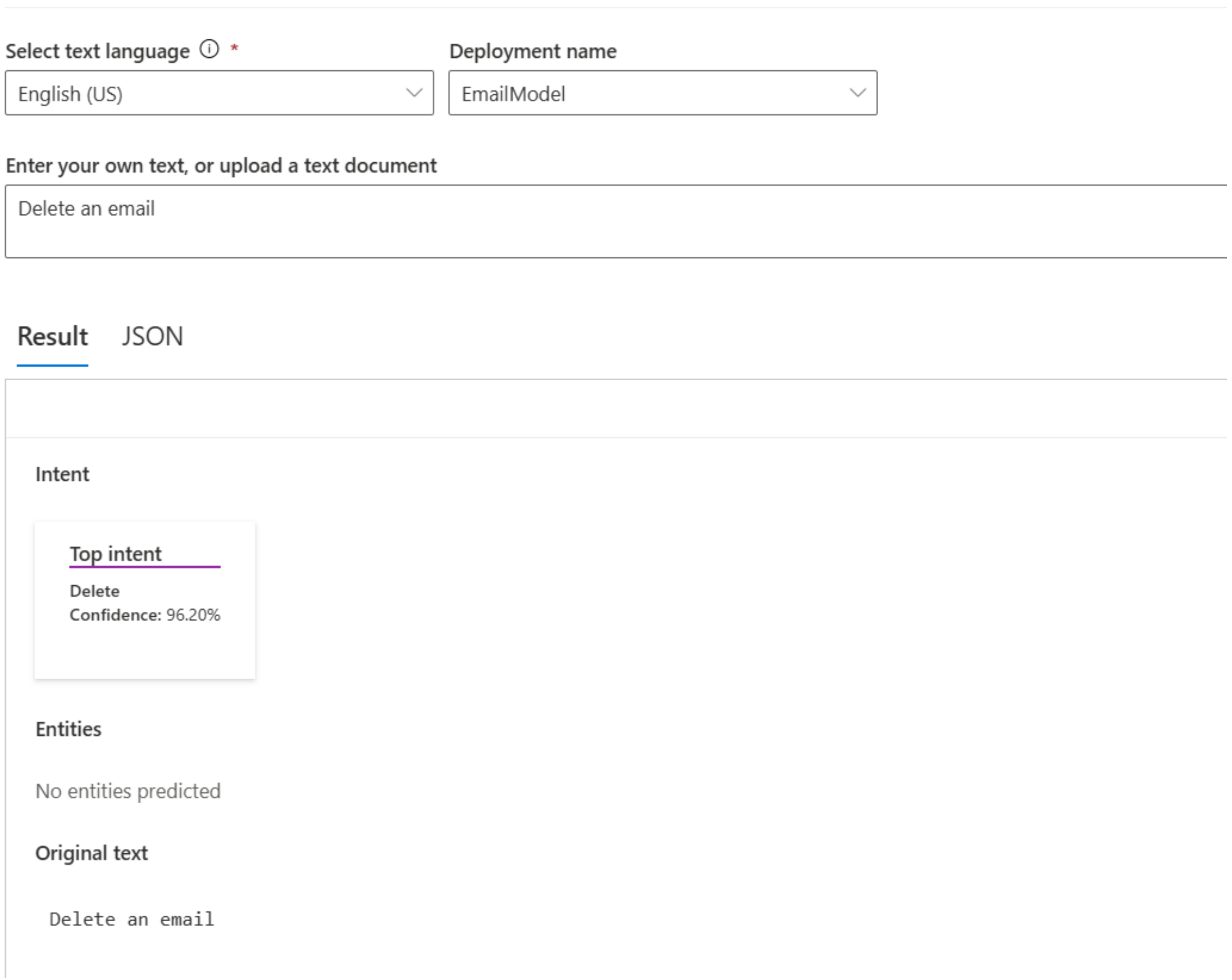
1 **Understand....**
- ✓ The principles of the conversational language understanding service
- ✓ The principles of training data in the conversational language understanding service

2 **Identify....**
- ✓ Use cases for this service

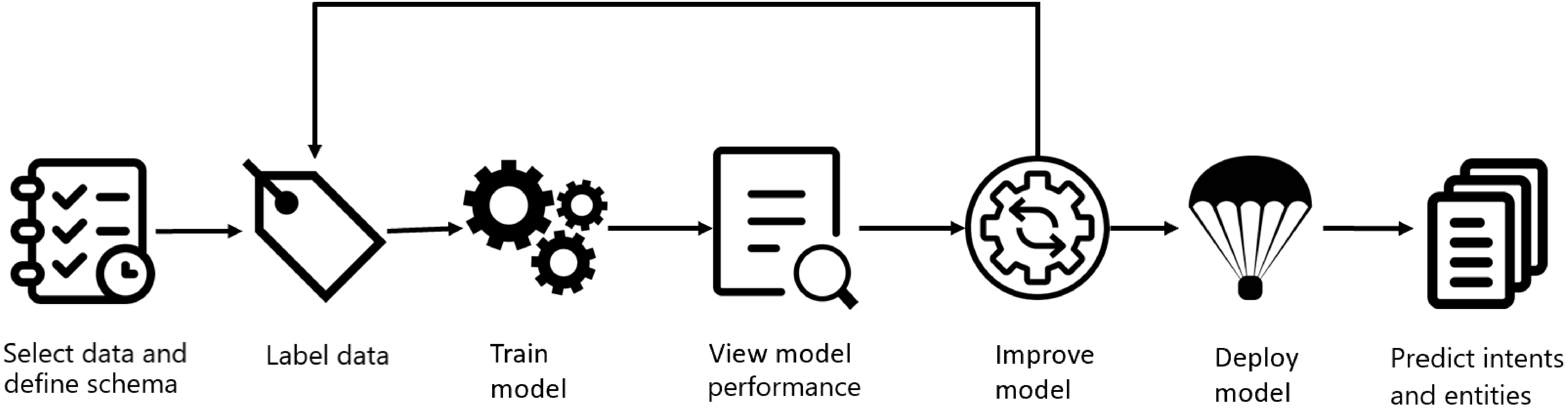
Conversational Language Understanding (CLU)

Example Applications

- End to end conversational bot
- Human assistant bots
- Command and control application
- Enterprise chat bot



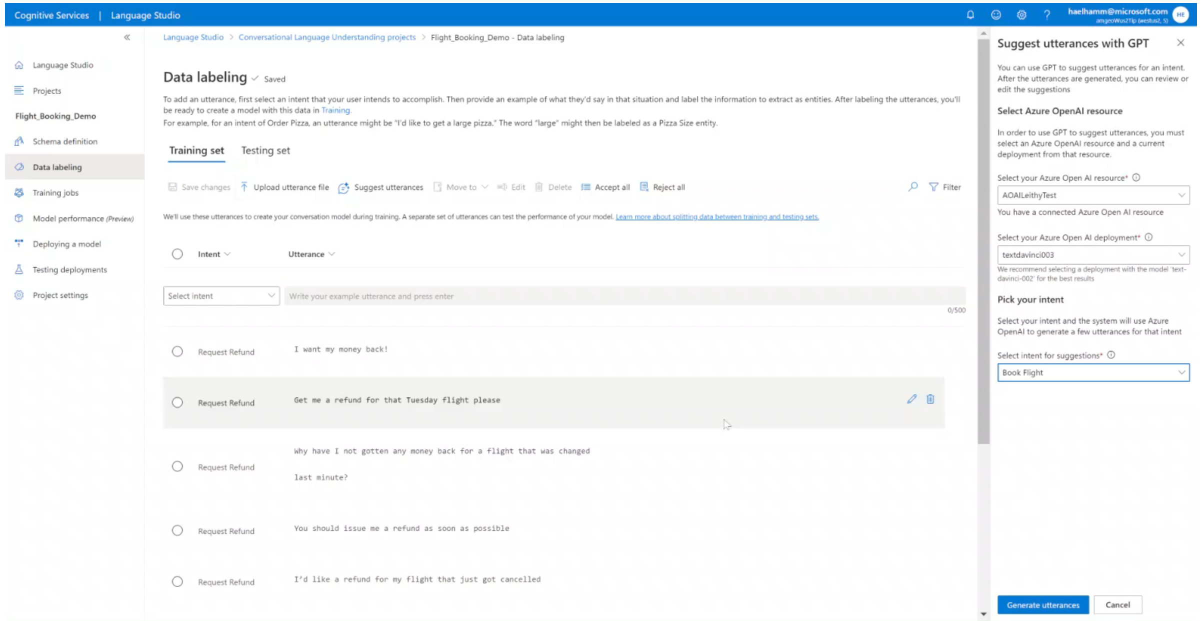
Conversational Language Understanding



CLU model training in Language Studio with AOAI

Define intents → Define entities for each intent → Training/Test intent identification and entity extraction → Deploy

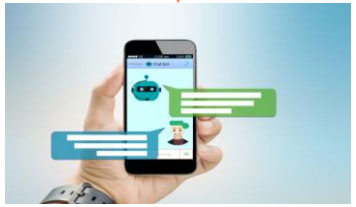
Supported Today:
GPT-powered training/testing data generation



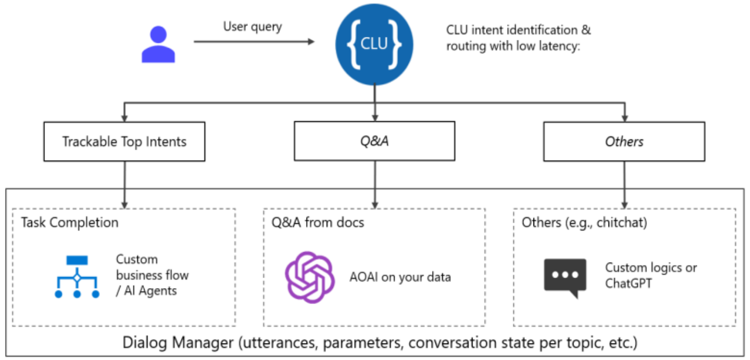
Why does Azure AI Language's CLU + AOAI Orchestration Service matter?

3. Azure AI Language's CLU & AOAI Orchestration – Key Customer Scenarios

CLU & AOAI Orchestration



Chat Intelligence



3. CLU & AOAI Orchestration – Key Highlights

Enhance Chat Intelligence by orchestrating CLU to track intent deflection rate with CLU's in-built **NER** model to improve chat experience with higher quality, cheaper cost and lower latency.

Trigger AI Agents to execute multiple custom business flows and complete tasks based on trackable **top intents** and user-defined **routing logic** to complete end-to-end **business process** tasks.

Orchestrating CLU with Azure OpenAI Service enables encoder model to extract entities from user utterances with higher accuracy

Intent classification in CLU ensures consistent intent schema while using LLMs to identify any new intents from long-tail topics

AI Translator

AI Translator Objectives

After completing this Learning, you will be able to:

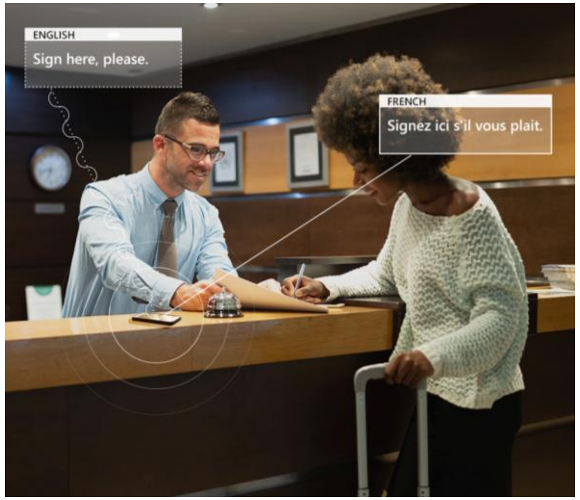
1	**Understand....** ✓ The principles of the AI Translator service ✓ The principles of information flow within the AI Translator service
2	**Identify....** ✓ Use cases for this service

Business Challenges

User-Generated Content / Social Media

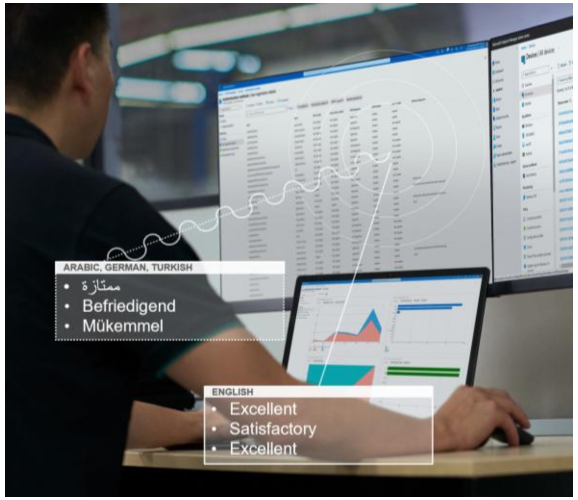
How to I reach all my customers in their preferred language

Localized Collateral



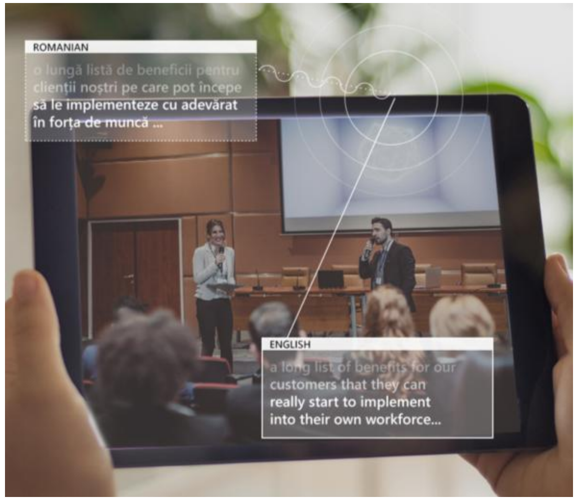
Are my documents localized making it easy to do business

E-Discovery



Content for business intelligence is in different languages across the web

Training Global Teams

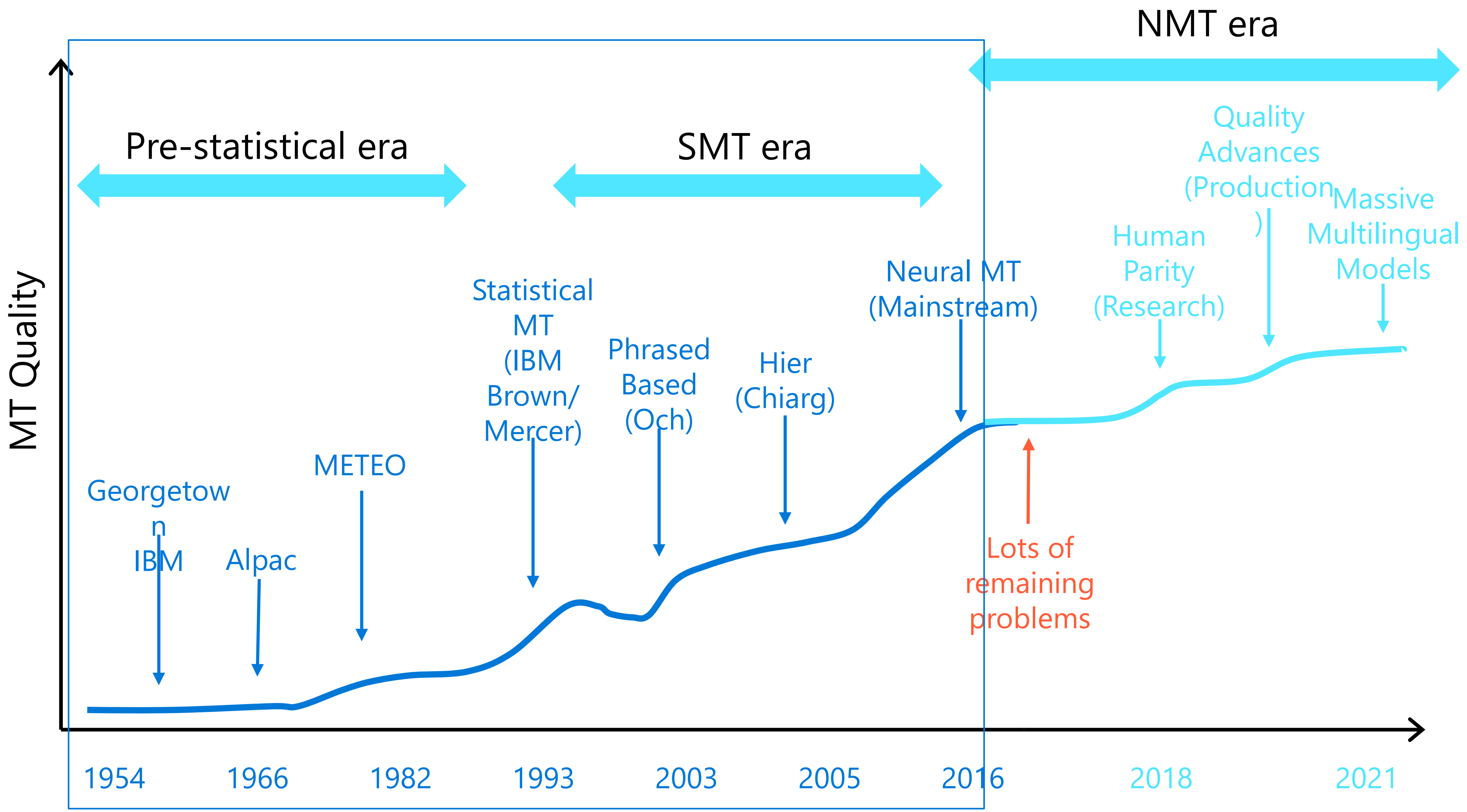


Not everyone has the same level of fluency in all languages

Translator Languages (133)

Afrikaans
Albanian
Amharic
Arabic
Armenian
Assamese
Azerbaijani
Bangla
Bashkir
Basque
Bhojpuri NEW!
Bosnian (Latin)
Bodo NEW!
Bulgarian
Cantonese (Traditional)
Catalan
Chinese (Literary)
Chinese Simplified
Chinese Traditional
chiShona
Croatian
Czech
Danish
Dari
Divehi
Dogri NEW!
Dutch
English
Estonian
Faroese
Fijian
Filipino
Finnish
French
French (Canada)
Galician
Georgian
German
Greek
Gujarati
Haitian Creole
Hausa
Hebrew
Hindi
Hmong Daw
Hungarian
Icelandic
Igbo
Indonesian
Inuinnaqtun
Inuktitut
Inuktitut (Latin)
Irish
Italian
Japanese
Kannada
Kashmiri NEW!
Kazakh
Khmer
Kinyarwanda
Klingon
Klingon (plqaD)
Konkani
Korean
Kurdish (Central)
Kurdish (Northern)
Kyrgyz
Lao
Latvian
Lingala
Lithuanian
Luganda
Macedonian
Maithili
Malagasy
Malay
Malayalam
Maltese
Māori
Marathi
Mongolian (Cyrillic)
Mongolian (Traditional)
Myanmar
Nepali
Norwegian
Nyanja
Odia
Pashto
Persian
Polish
Portuguese (Brazil)
Portuguese (Portugal)
Punjabi
Queretaro Otomi
Romanian
Rundi
Russian
Samoan
Serbian (Cyrillic)
Serbian (Latin)
Sesotho
Sesotho sa Leboa
Setswana
Sindhi
Sinhala
Slovak
Slovenian
Somali
Sorbian (Lower)
Sorbian (Upper)
Spanish
Swahili
Swedish
Tahitian
Tamil
Tatar
Telugu
Thai
Tibetan
Tigrinya
Tongan
Turkish
Turkmen
Ukrainian
Urdu
Uyghur
Uzbek (Latin)
Vietnamese
Welsh
Xhosa
Yoruba
Yucatec Maya
Zulu

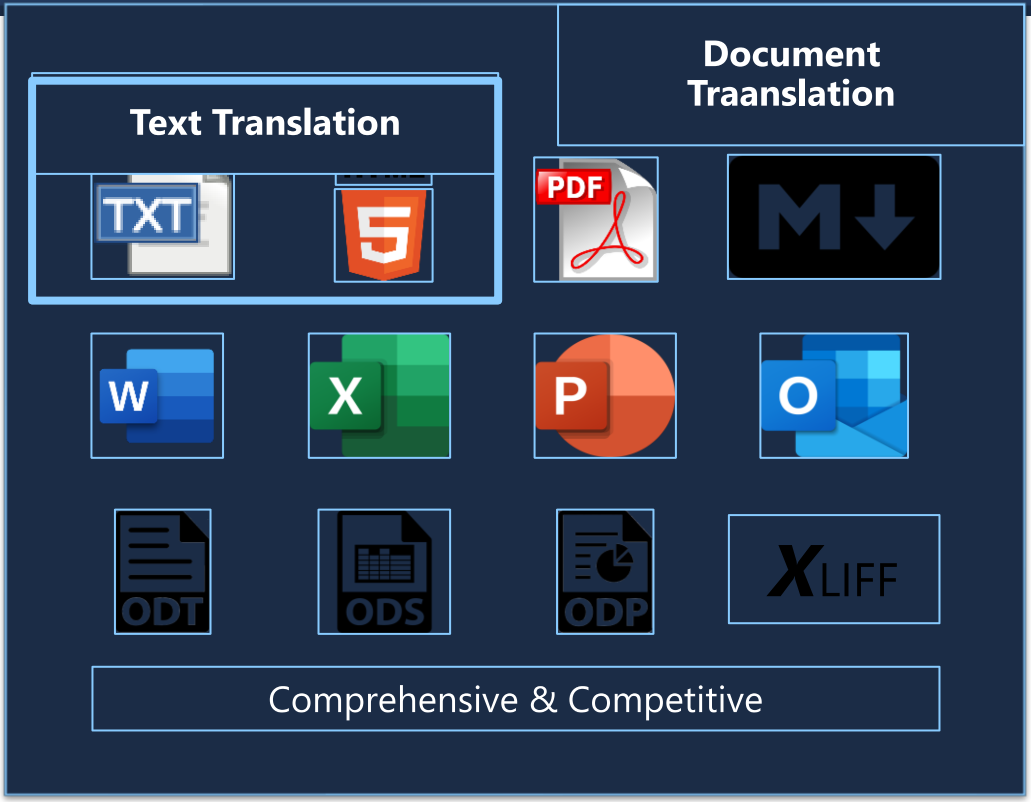
Progress in Machine Translation (MT)



Chris Manning 2016

Translator Overview

- Translate Text
- Translate Documents
- Translate scanned PDF documents (Image)
- Translate Speech
- Customization
  - Dictionary & Model
- Security & Compliance
- Cloud & On-premises container offering



Regulatory Compliance

CSA | FedRAMP | GDPR | HIPAA | HITRUST | ISO | PCI | SOC | US DoD

Where is my request processed?

	Region in which resource is created	Service endpoint	Request processed by data center
Translator (text)	Any	**Global (recommended)**: api.cognitive.microsofttranslator.com	Closest available datacenter
		Americas: api-nam.cognitive.microsofttranslator.com	East US, South Central US, West Central US, and West US 2
		Europe: api-eur.cognitive.microsofttranslator.com	North Europe and West Europe
		Asia Pacific: api-apc.cognitive.microsofttranslator.com	Korea South, Japan East, Southeast Asia, and Australia East
Translator (document)	Global and any region in Americas	**Custom**: <name_of_your_resource>.cognitiveservices.azure.com/translator/text/batch/v1.0	East US and West US 2
	Any region in Europe		North Europe and West Europe
	Any region in Asia Pacific		Southeast Asia and Australia East

Document Translation overview

Translates rich documents retaining structure and layout

Translates documents into multiple languages

Translates batch of large documents

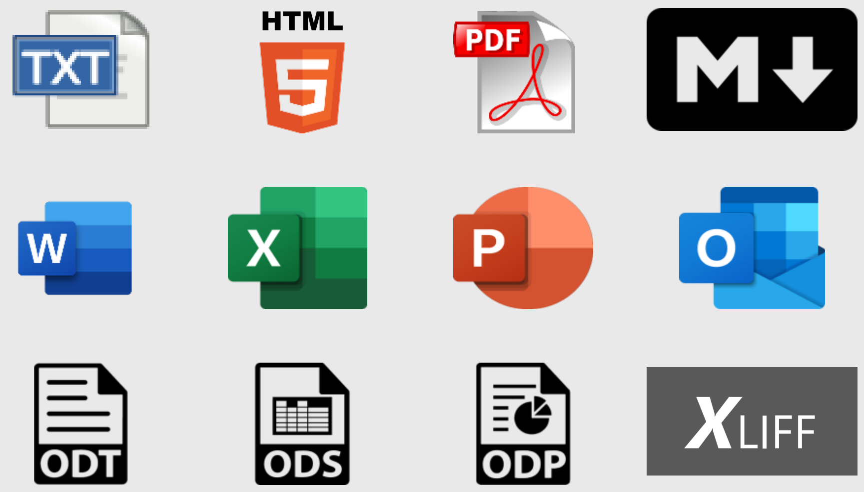
Translates multi-lingual documents

Uses general & custom models

Applies custom glossaries

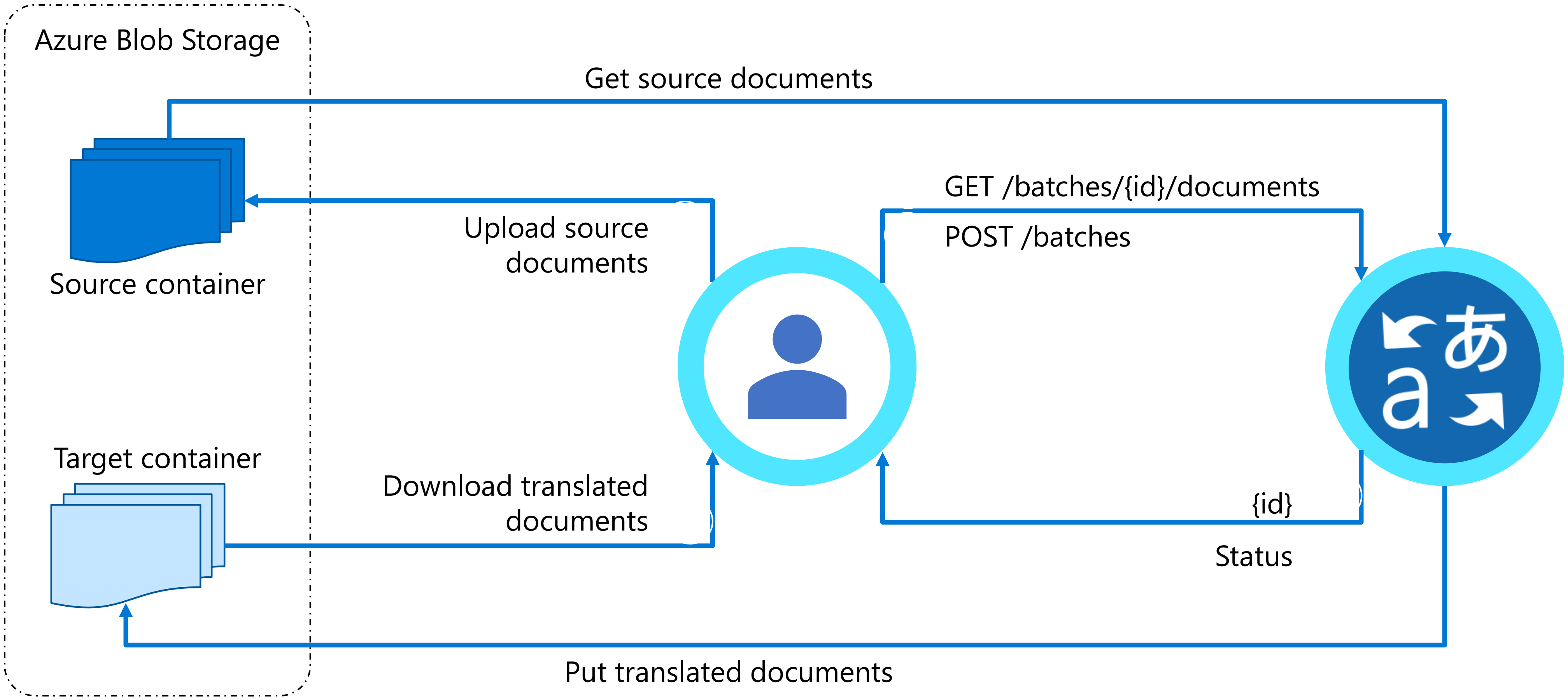
Azure AD authentication

Secured access to storage using managed identity



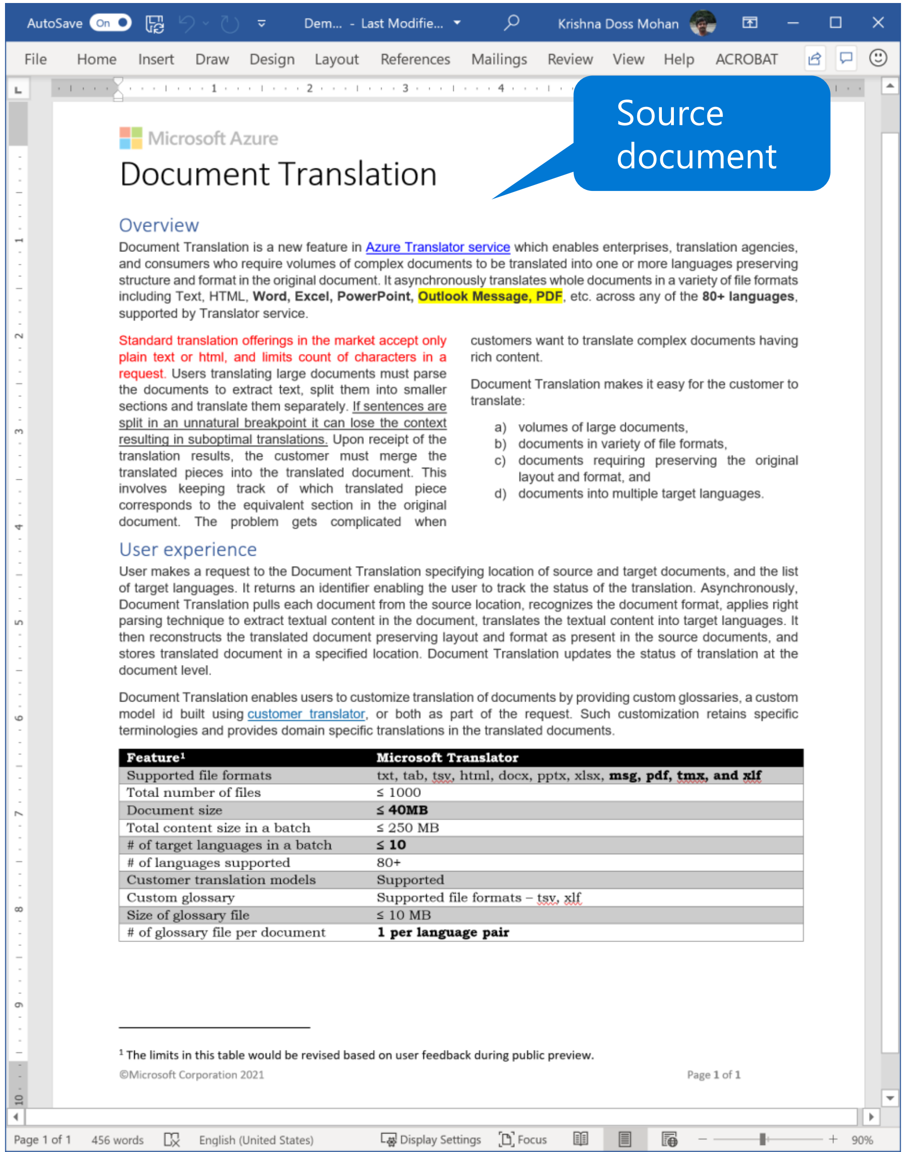
Comprehensive & Competitive offering in the market covering all popular document formats

Document Translation simple flow



Document Translation

Translate at scale
and preserve
format



Microsoft Azure

Document Translation

Overview

Document Translation is a new feature in Azure Translator service which enables enterprises, translation agencies, and consumers who require volumes of complex documents to be translated into one or more languages preserving structure and format in the original document. It asynchronously translates whole documents in a variety of file formats including Text, HTML, **Word, Excel, PowerPoint, Outlook Message, PDF**, etc. across any of the **80+ languages**, supported by Translator service.

Standard translation offerings in the market accept only plain text or html, and limits count of characters in a request. Users translating large documents must parse the documents to extract text, split them into smaller sections and translate them separately. If sentences are split in an unnatural breakpoint it can lose the context resulting in suboptimal translations. Upon receipt of the translation results, the customer must merge the translated pieces into the translated document. This involves keeping track of which translated piece corresponds to the equivalent section in the original document. The problem gets complicated when customers want to translate complex documents having rich content.

Document Translation makes it easy for the customer to translate:

a) volumes of large documents,
b) documents in variety of file formats,
c) documents requiring preserving the original layout and format, and
d) documents into multiple target languages.

User experience

User makes a request to the Document Translation specifying location of source and target documents, and the list of target languages. It returns an identifier enabling the user to track the status of the translation. Asynchronously, Document Translation pulls each document from the source location, recognizes the document format, applies right parsing technique to extract textual content in the document, translates the textual content into target languages. It then reconstructs the translated document preserving layout and format as present in the source documents, and stores translated document in a specified location. Document Translation updates the status of translation at the document level.

Document Translation enables users to customize translation of documents by providing custom glossaries, a custom model id built using customer translator, or both as part of the request. Such customization retains specific terminologies and provides domain specific translations in the translated documents.

Feature[1]	Microsoft Translator
Supported file formats	txt, tab, tsv, html, docx, pptx, xlsx, **msg, pdf, tmx, and xlf**
Total number of files	≤ 1000
Document size	**≤ 40MB**
Total content size in a batch	≤ 250 MB
# of target languages in a batch	**≤ 10**
# of languages supported	80+
Customer translation models	Supported
Custom glossary	Supported file formats – tsv, xlf
Size of glossary file	≤ 10 MB
# of glossary file per document	**1 per language pair**

[1] The limits in this table would be revised based on user feedback during public preview.

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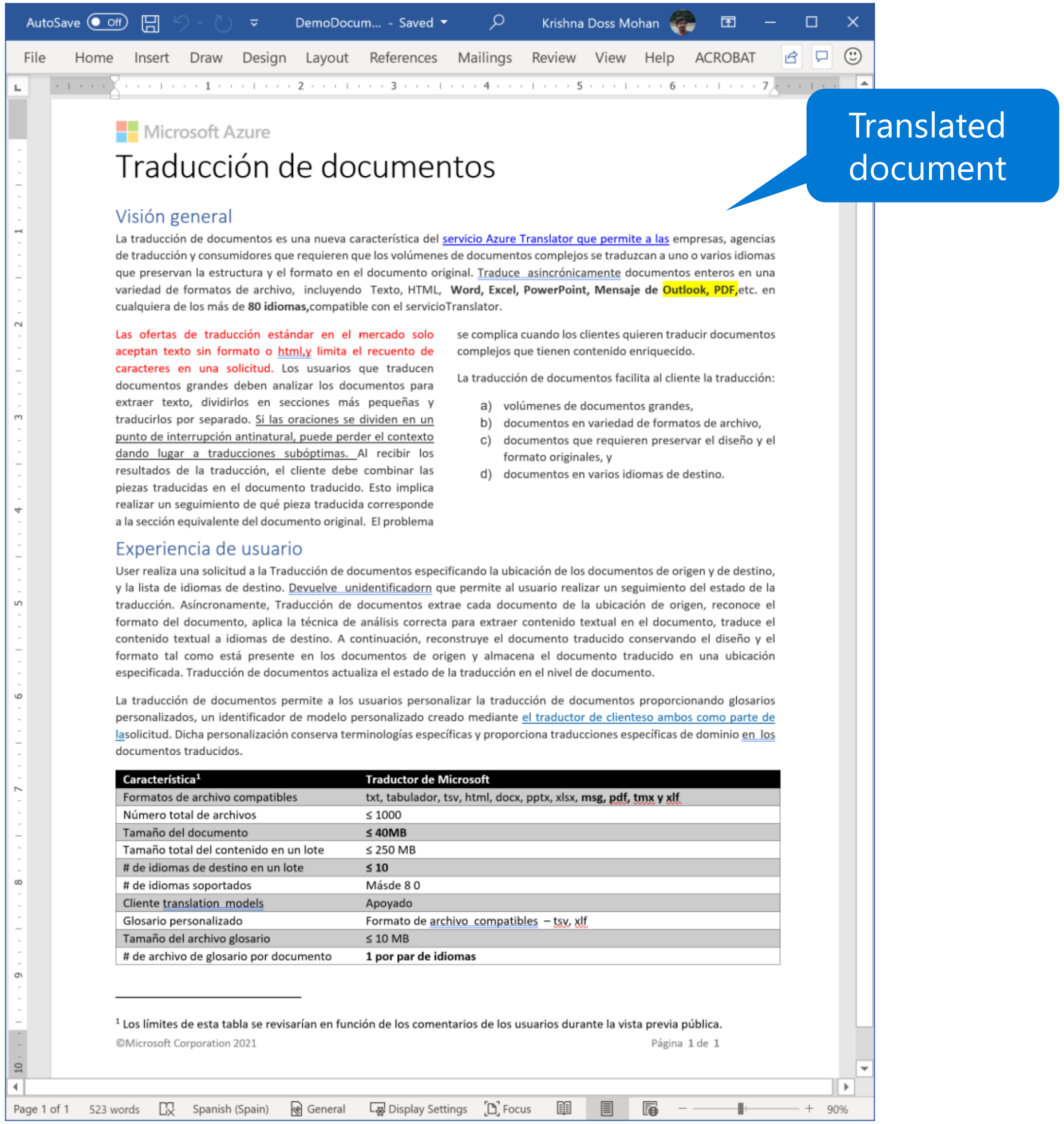

Microsoft Azure

Traducción de documentos

Visión general

La traducción de documentos es una nueva característica del servicio Azure Translator que permite a las empresas, agencias de traducción y consumidores que requieren que los volúmenes de documentos complejos se traduzcan a uno o varios idiomas que preservan la estructura y el formato en el documento original. Traduce asincrónicamente documentos enteros en una variedad de formatos de archivo, incluyendo Texto, HTML, **Word, Excel, PowerPoint, Mensaje de Outlook, PDF,**etc. en cualquiera de los más de **80 idiomas,**compatible con el servicioTranslator.

Las ofertas de traducción estándar en el mercado solo aceptan texto sin formato o html,y limita el recuento de caracteres en una solicitud. Los usuarios que traducen documentos grandes deben analizar los documentos para extraer texto, dividirlos en secciones más pequeñas y traducirlos por separado. Si las oraciones se dividen en un punto de interrupción antinatural, puede perder el contexto dando lugar a traducciones subóptimas. Al recibir los resultados de la traducción, el cliente debe combinar las piezas traducidas en el documento traducido. Esto implica realizar un seguimiento de qué pieza traducida corresponde a la sección equivalente del documento original. El problema se complica cuando los clientes quieren traducir documentos complejos que tienen contenido enriquecido.

La traducción de documentos facilita al cliente la traducción:

a) volúmenes de documentos grandes,
b) documentos en variedad de formatos de archivo,
c) documentos que requieren preservar el diseño y el formato originales, y
d) documentos en varios idiomas de destino.

Experiencia de usuario

User realiza una solicitud a la Traducción de documentos especificando la ubicación de los documentos de origen y de destino, y la lista de idiomas de destino. Devuelve unidentificadorn que permite al usuario realizar un seguimiento del estado de la traducción. Asíncronamente, Traducción de documentos extrae cada documento de la ubicación de origen, reconoce el formato del documento, aplica la técnica de análisis correcta para extraer contenido textual en el documento, traduce el contenido textual a idiomas de destino. A continuación, reconstruye el documento traducido conservando el diseño y el formato tal como está presente en los documentos de origen y almacena el documento traducido en una ubicación especificada. Traducción de documentos actualiza el estado de la traducción en el nivel de documento.

La traducción de documentos permite a los usuarios personalizar la traducción de documentos proporcionando glosarios personalizados, un identificador de modelo personalizado creado mediante el traductor de clienteso ambos como parte de lasolicitud. Dicha personalización conserva terminologías específicas y proporciona traducciones específicas de dominio en los documentos traducidos.

Característica[1]	Traductor de Microsoft
Formatos de archivo compatibles	txt, tabulador, tsv, html, docx, pptx, xlsx, **msg, pdf, tmx y xlf**
Número total de archivos	≤ 1000
Tamaño del documento	**≤ 40MB**
Tamaño total del contenido en un lote	≤ 250 MB
# de idiomas de destino en un lote	**≤ 10**
# de idiomas soportados	Másde 8 0
Cliente translation models	Apoyado
Glosario personalizado	Formato de archivo compatibles – tsv, xlf
Tamaño del archivo glosario	≤ 10 MB
# de archivo de glosario por documento	**1 por par de idiomas**

[1] Los límites de esta tabla se revisarían en función de los comentarios de los usuarios durante la vista previa pública.

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Scanned PDF translation

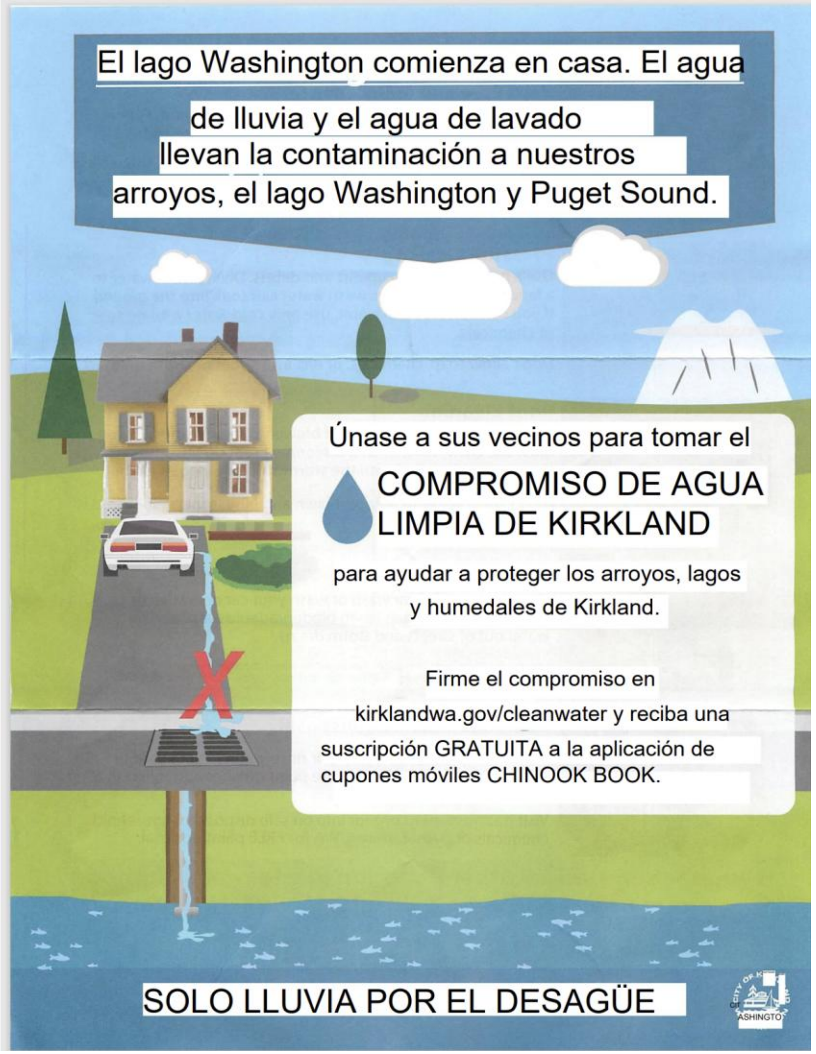
Document translation now can translate PDF documents having scanned content

Document translation service internally calls OCR, extracts text, translates and reconstructs a digital PDF document

- Font formatting like bold, italics, underline, highlights, etc. aren't retained
- Currently content written right-to-left and top-to-bottom are not supported

Support of scanned PDF translation comes at no additional cost to the customer.

Translation Example



Translation Example

Translator API limits

Max request size: 10K characters

Character limits per hour:

Tier	Character limit
F0	2 million characters per hour
S1	40 million characters per hour
S2 / C2	40 million characters per hour
S3 / C3	120 million characters per hour
S4 / C4	200 million characters per hour

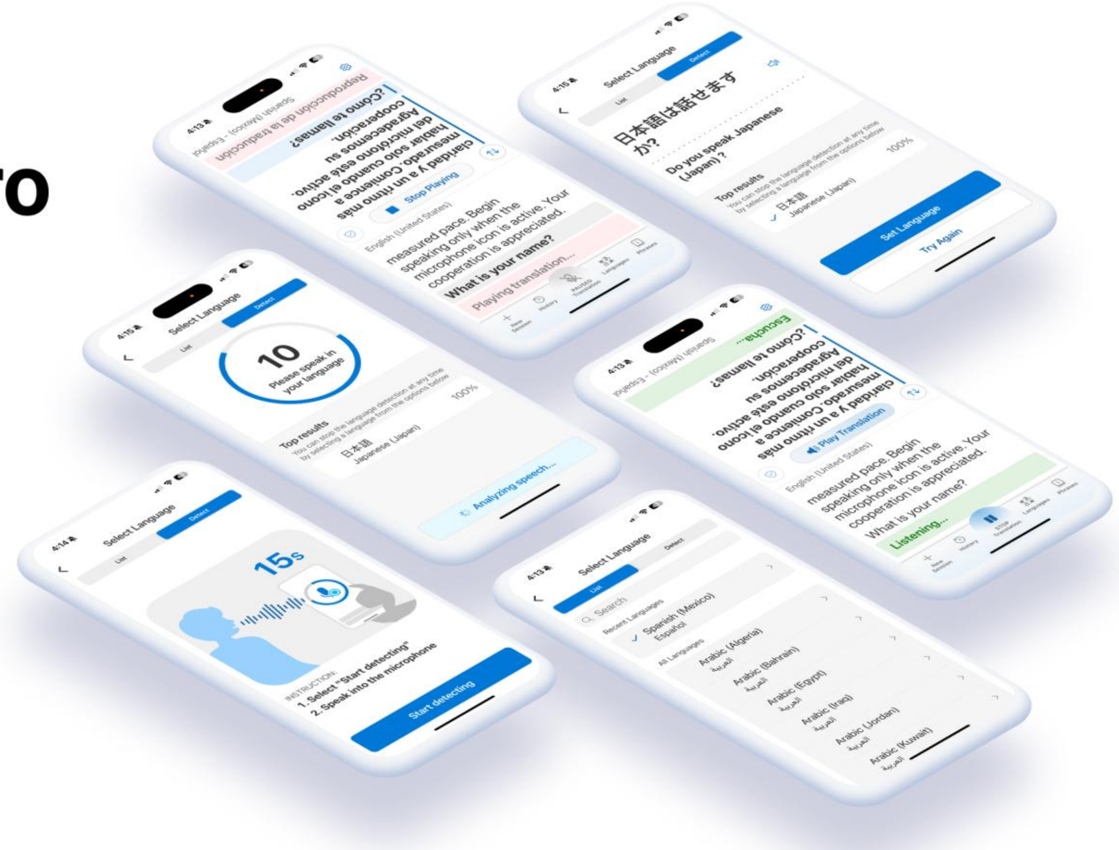
Custom translation limited to 1,800 characters/second

Average latency: 150 milliseconds to 300 milliseconds

Max latency: 15 seconds for standard models, 120 seconds for custom models

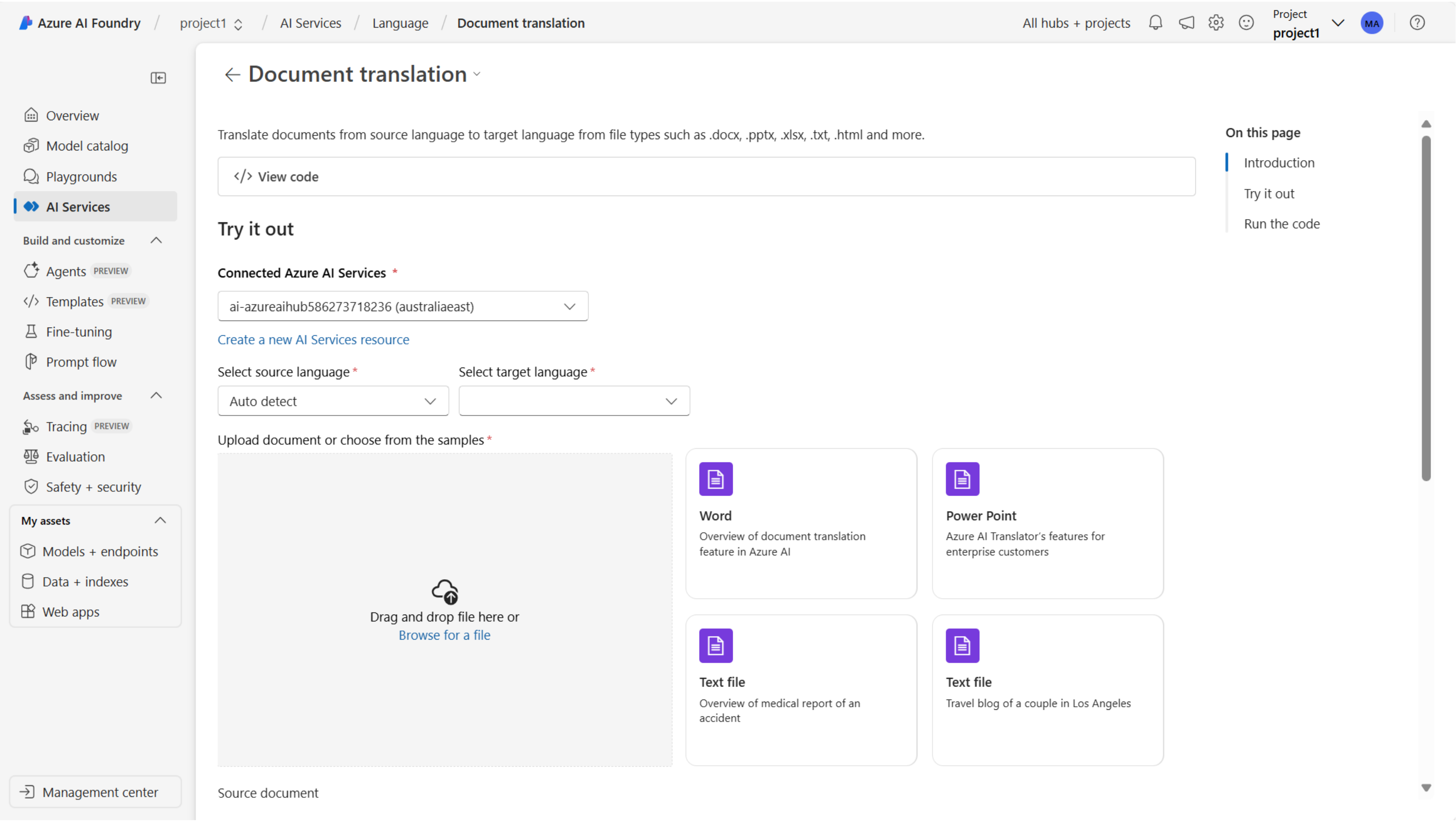
Microsoft Translator Pro

Microsoft
Translator Pro



AI Translator Demonstration

AI Translator



Which language service should I use?

Which Language service feature should I use?

What do you want to do?	Document format	Recommended Solution	Is this solution customizable?*
Detect and/or redact sensitive information such as PII and PHI.	Unstructured text, transcribed conversations	PII detection	
Extract categories of information without creating a custom model.	Unstructured text	The preconfigured NER feature	
Extract categories of information using a model specific to your data.	Unstructured text	Custom NER	✓
Extract main topics and important phrases.	Unstructured text	Key phrase extraction	
Determine the sentiment and opinions expressed in text.	Unstructured text	Sentiment analysis and opinion mining	
Summarize long chunks of text or conversations.	Unstructured text, transcribed conversations.	Summarization	
Disambiguate entities and get links to Wikipedia.	Unstructured text	Entity linking	
Classify documents into one or more categories.	Unstructured text	Custom text classification	✓
Extract medical information from clinical/medical documents, without building a model.	Unstructured text	Text analytics for health	
Build a conversational application that responds to user inputs.	Unstructured user inputs	Question answering	✓
Detect the language that a text was written in.	Unstructured text	Language detection	
Predict the intention of user inputs and extract information from them.	Unstructured user inputs	Conversational language understanding	✓
Connect apps from conversational language understanding, LUIS, and question answering.	Unstructured user inputs	Orchestration workflow	✓

Should I choose Azure AI Language or Azure Open AI?

AI Language

- Targeted focus and capabilities
- Deterministic and repeatable
- Available as a managed service or containerized (in some cases) to run locally/securely
- Excels in scale and low latency
- Customizable
- Returns a confidence score as part of the response
- No prompting required

Azure OpenAI

- Broad capabilities focused on generation and natural language interactions
- Not inherently deterministic
- Hosting/Fine tuning requires large amounts of compute so generally used as a managed service
- Highly capable in language tasks

Responsible Use

Responsible AI Objectives

After completing this Learning, you will be able to:

1	**Understand….** ✓ The principles of Responsible AI ✓ How to access transparency notes for AI Language services
2	**Identify….** ✓ Monitoring and Logging options for your AI Language tools

Microsoft's Responsible AI principles

Fairness

Reliability & Safety

Privacy & Security

Inclusiveness

Transparency

Accountability

Learn more at aka.ms/RAI

Responsible AI / Transparency Notes

- Transparency Note for Azure AI Language
- Transparency note for Named Entity Recognition and Personally Identifying Information
- Transparency note for the health feature
- Transparency note for Key Phrase Extraction
- Transparency note for Language Detection
- Transparency note for Question answering
- Transparency note for Summarization
- Transparency note for Sentiment Analysis
- Azure AI Translator Transparency Note

Data, Privacy and Security

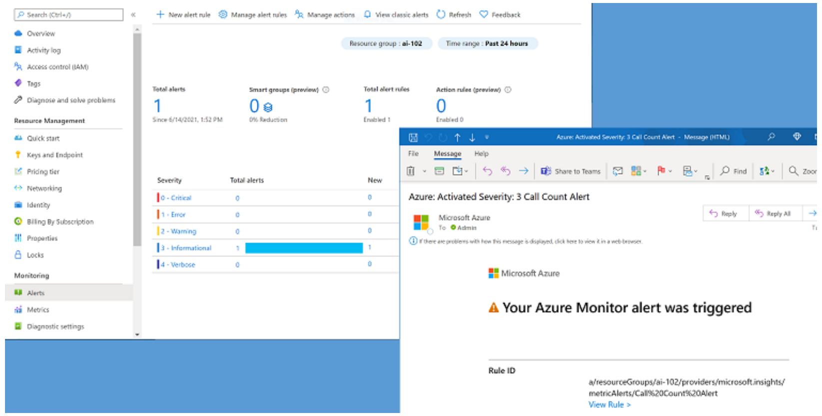
What data does Azure AI Language process and how does it process it?

- Azure AI Language processes text data that is sent by the customer
- All results are sent back to the customer in the API response
- Azure AI Language uses aggregate telemetry
- Azure AI Language doesn't store or process customer data outside the region
- Azure AI Language encrypts all content, including customer data, at rest.

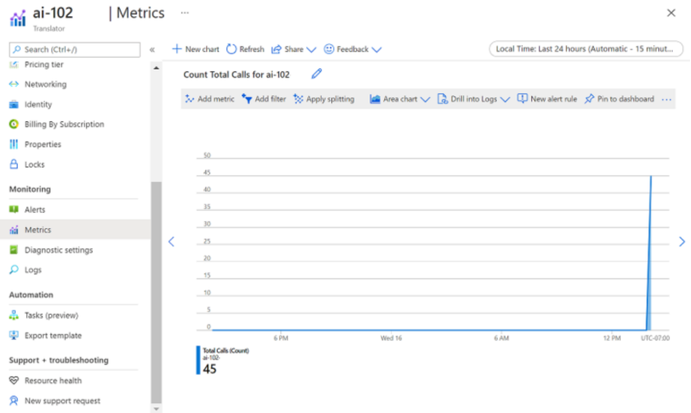
How is data retained and what customer controls are available?

- Data sent in synchronous or asynchronous calls may be temporarily stored by Azure AI Language for up to 48 hours only and is purged thereafter.
- To prevent this temporary storage of input data, the LoggingOptOut query parameter can be set accordingly.

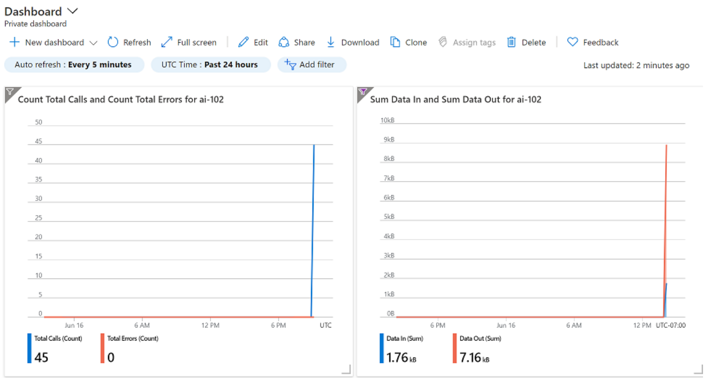
Monitoring and Logging



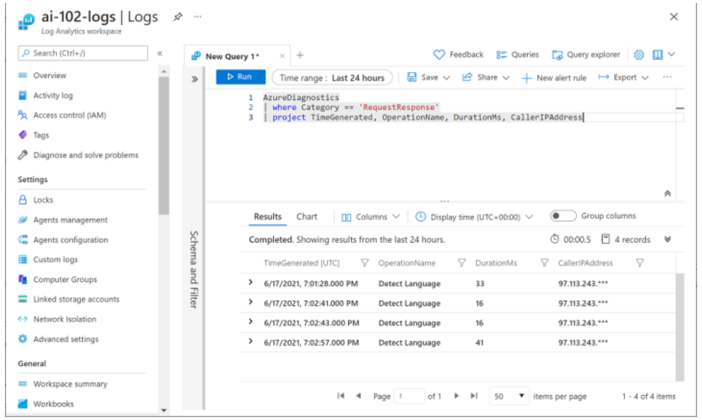
Create Cost Alerts



View Metrics



Create a Dashboard



Manage Diagnostic Logging

Containerization, Integration, Examples and Task Agents

Containerization Objectives

After completing this Learning, you will be able to:

1	**Understand....** ✓ The principles of containerized language services ✓ Sample architectures and use cases for the AI Language service
2	**Identify....** ✓ Where Task Agents could make use of the AI Language service

Microsoft

Containerized language services


Microsoft
Azure

Azure AI task-specific models and capabilities

Speech

Enable every device to hear and speak

- Speech to text
- Text to speech
- Custom Neural Voice
- Speech Translation
- Speaker Diarization
- Language Identification

Translator

Understand any language

- Text translation
- Document translation
- Microsoft Translator Pro (preview)
- Embedded translation

Language

Extract information & process intents

- Summarization
- Entity extraction (PII/PHI)
- Conversational AI
- Text analytics
- Sentiment mining

Content Understanding

Understand multimodal contents

- Document Intelligence
- Image analysis
- Video analysis
- Audio analysis
- Facial recognition

Responsible AI

Ensure AI use remains safe

- Content safety filters
- Safety evaluations

AI task-specific models available in

- Microsoft Public Cloud
- Microsoft Government Cloud
- Connected Containers
- Disconnected Containers

Regulatory Compliance

CSA cloud security alliance · FedRAMP · GDPR · HIPAA COMPLIANT · HITRUST · ISO · PCI DSS COMPLIANT · AICPA SOC · Department of Defense

Customer pain points

Data security & privacy concerns

Unable to upload all audio/text data to the cloud

Low bandwidth or intermittent connectivity

Azure AI Container Capabilities

Immutable infrastructure | Adapt to changes and avoid configuration drift

Control over data | Choose where data is processed with drive data, management, identity & security consistency

Control over model updates | Flexible versioning and control over when to deploy in your environments

Portable architecture | Create portable app architecture to deploy on Azure, on-prem, and edge endpoints

High throughput, Low latency | Run apps physically close to data and you control the throughput

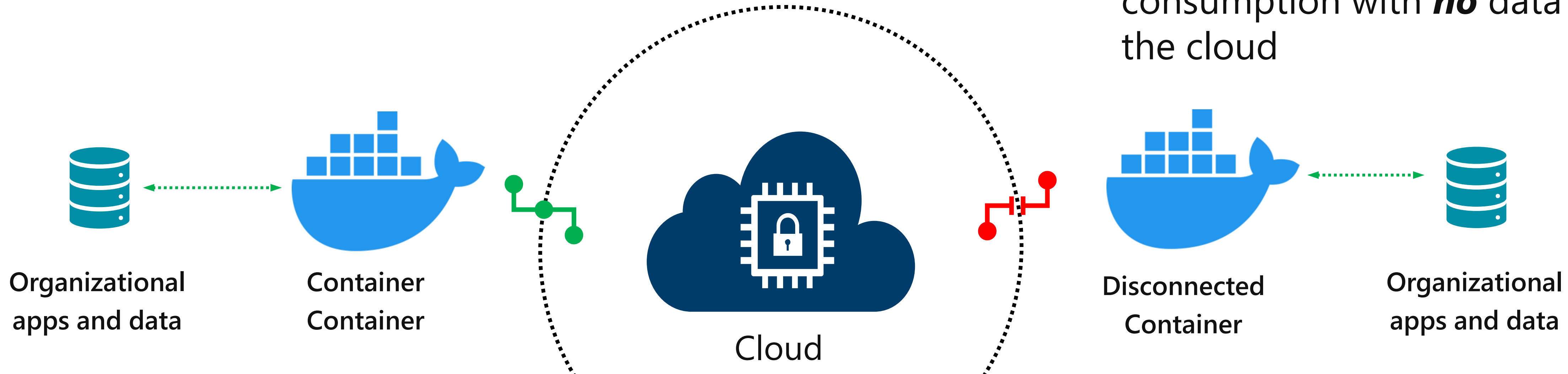
Scalability | Build your own apps on a scalable cluster foundation based on your business requirements

AI Container environment options

Pay-as-you-go consumption model where ***only*** billing data is sent to the cloud

- Connected containers
- Disconnected containers

Pay up front for a years' worth of consumption with ***no*** data sent to the cloud



Designed for specific security and data governance requirements where billing can be sent

Designed for environments with no cloud connectivity

AI task-specific models available in Containers

Speech

Speech to Text

Text to Speech

Neural Speech to Text

Custom Text to Speech*

Speech Language Identification (Preview)*

Speaker Diarization*

Translataor

Text Translation

Document Translation

Transliteration

Language

Language Detection

Key Phrase Extraction

Sentiment Analysis

PII Redaction

Summarization

CLU

NER

Custom NER*

Text Analytics for Health*

Content Understanding

Document Intelligence includes OCR - designed for documents

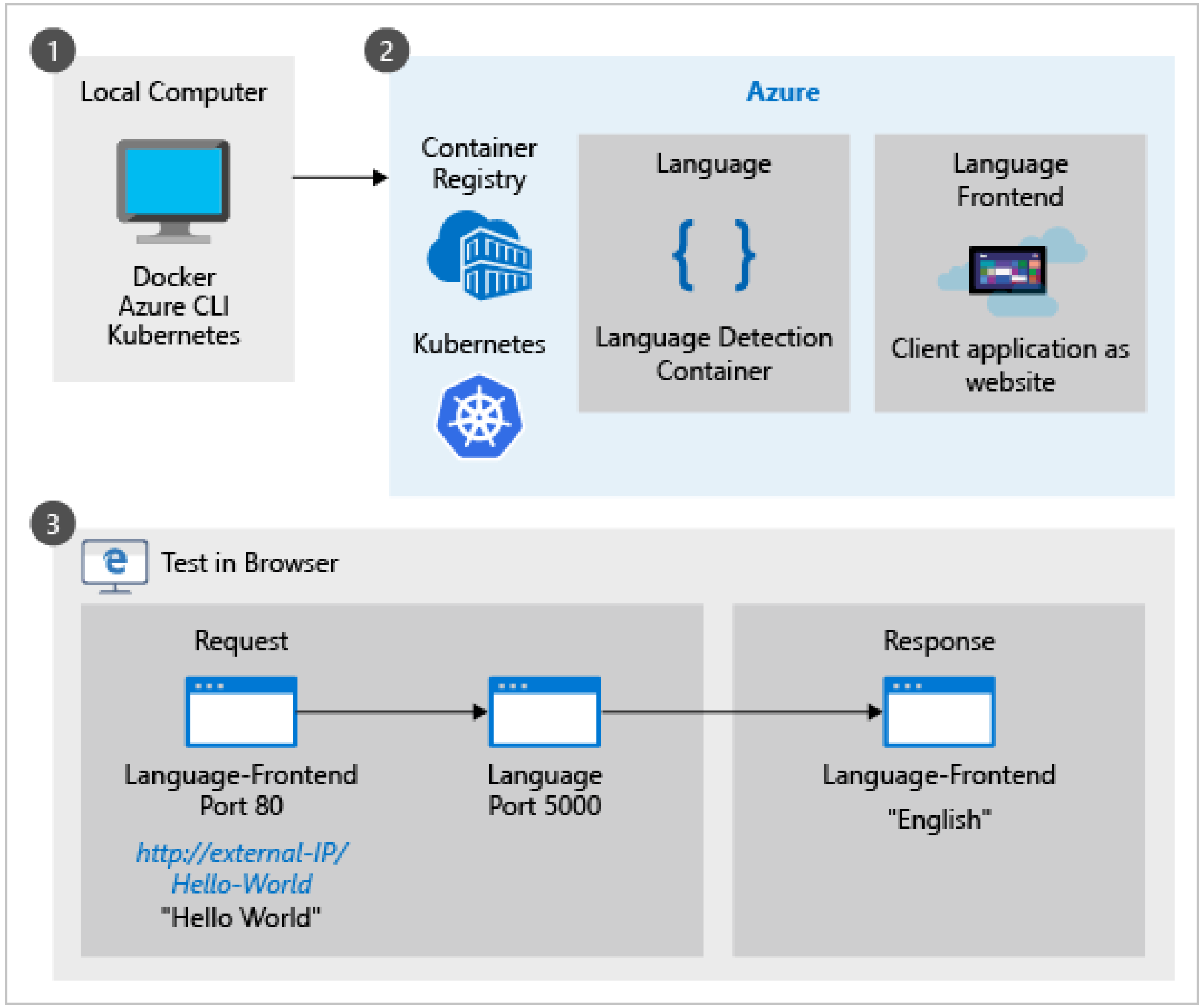
Computer vision OCR optimized for images*

Responsible AI

Content Safety (limited access preview)

* **Available** *only* in connected containers

Deploy a language detection container to Azure Kubernetes Service

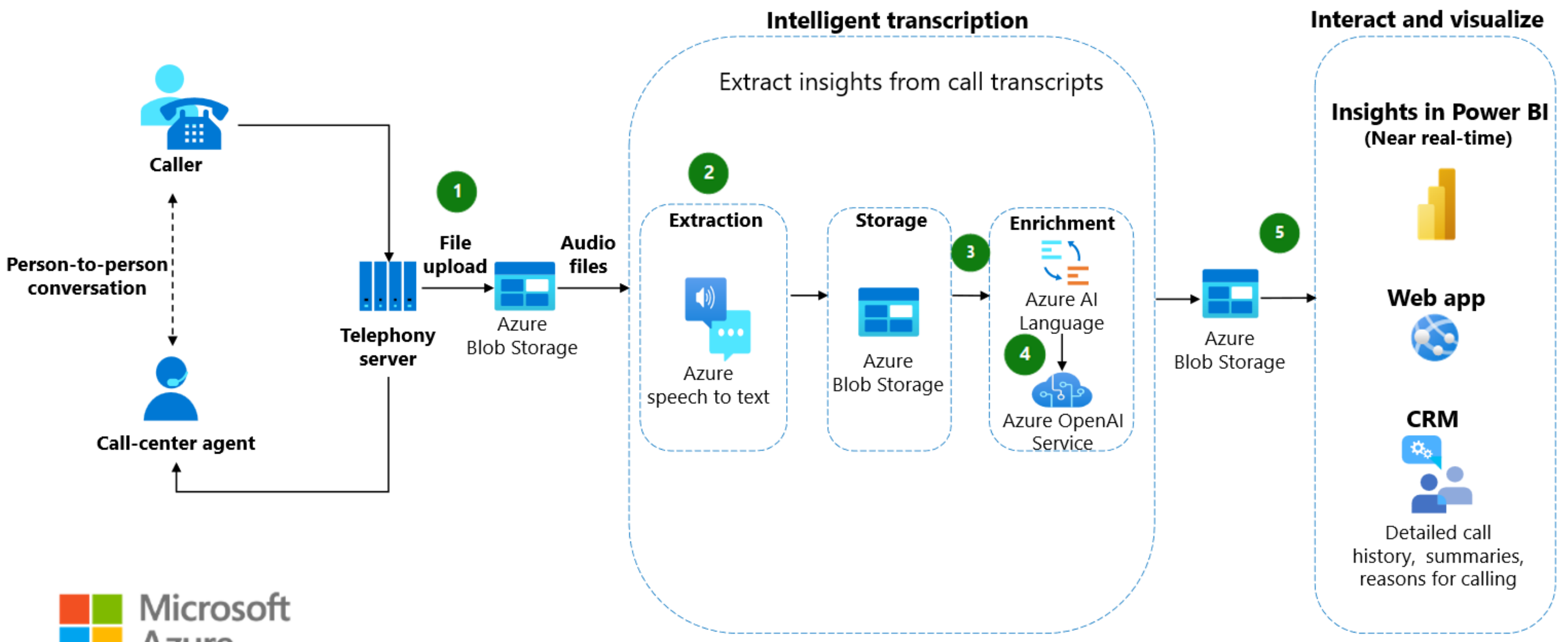


https://learn.microsoft.com/en-us/azure/ai-services/containers/azure-kubernetes-recipe

Example: Accelerator for disconnected AI Services

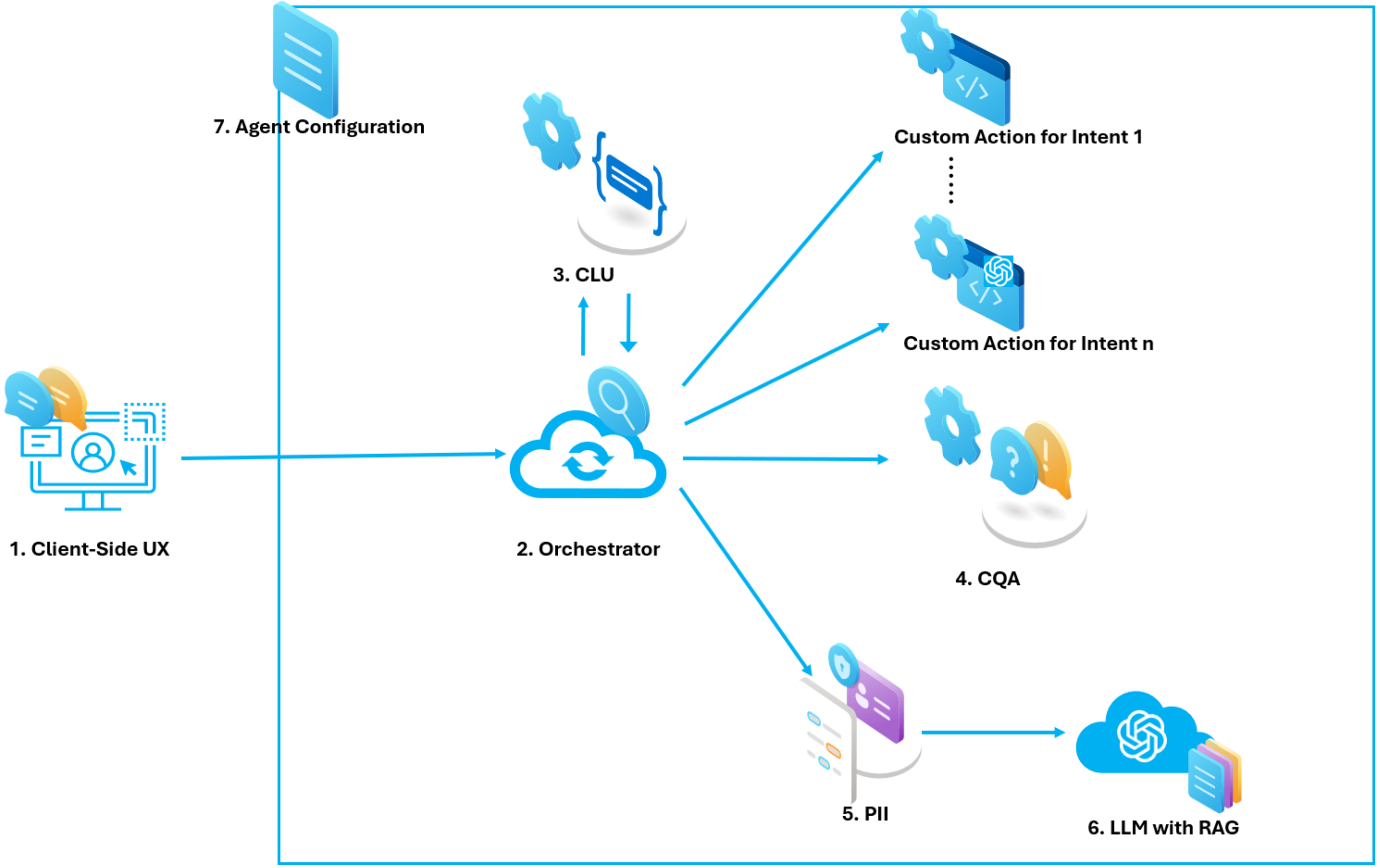
Sample Architectures / Examples

Extract and analyze call center data

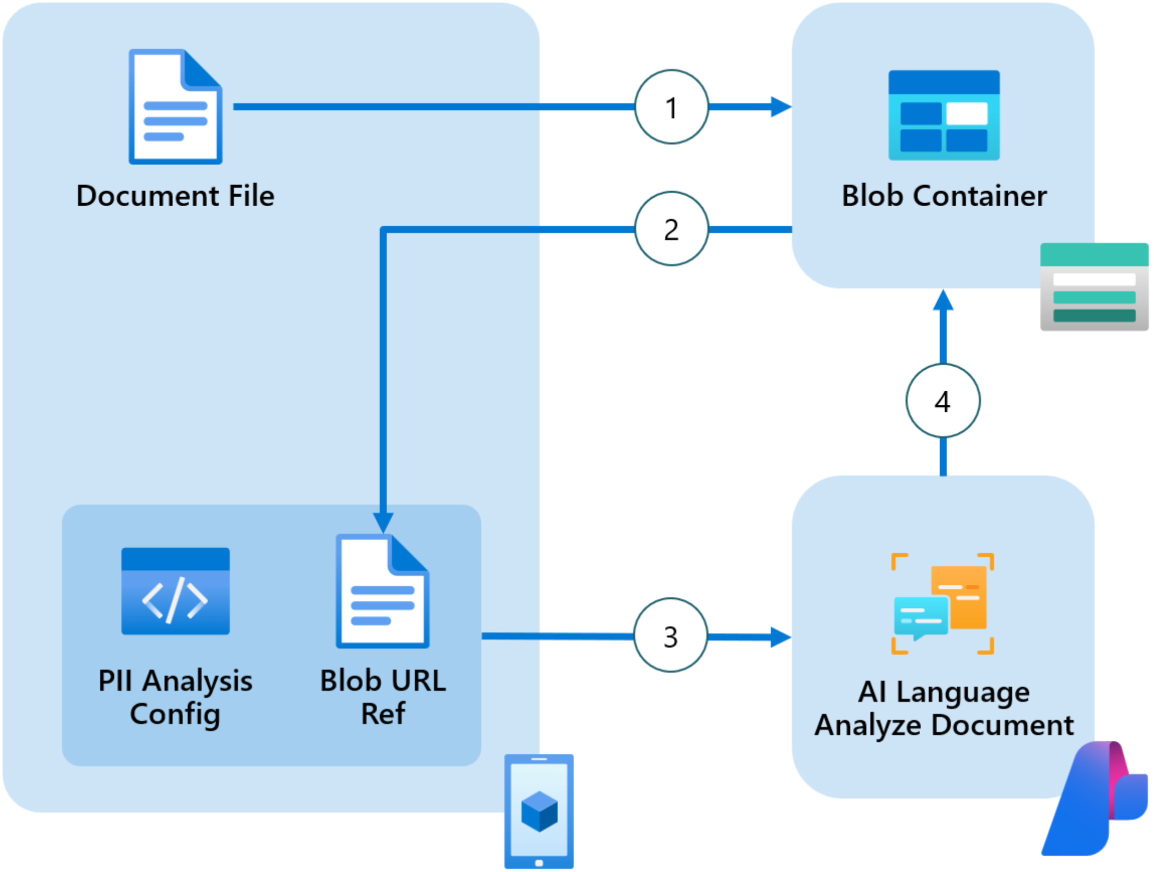


Azure Language OpenAI Conversational Agent Accelerator

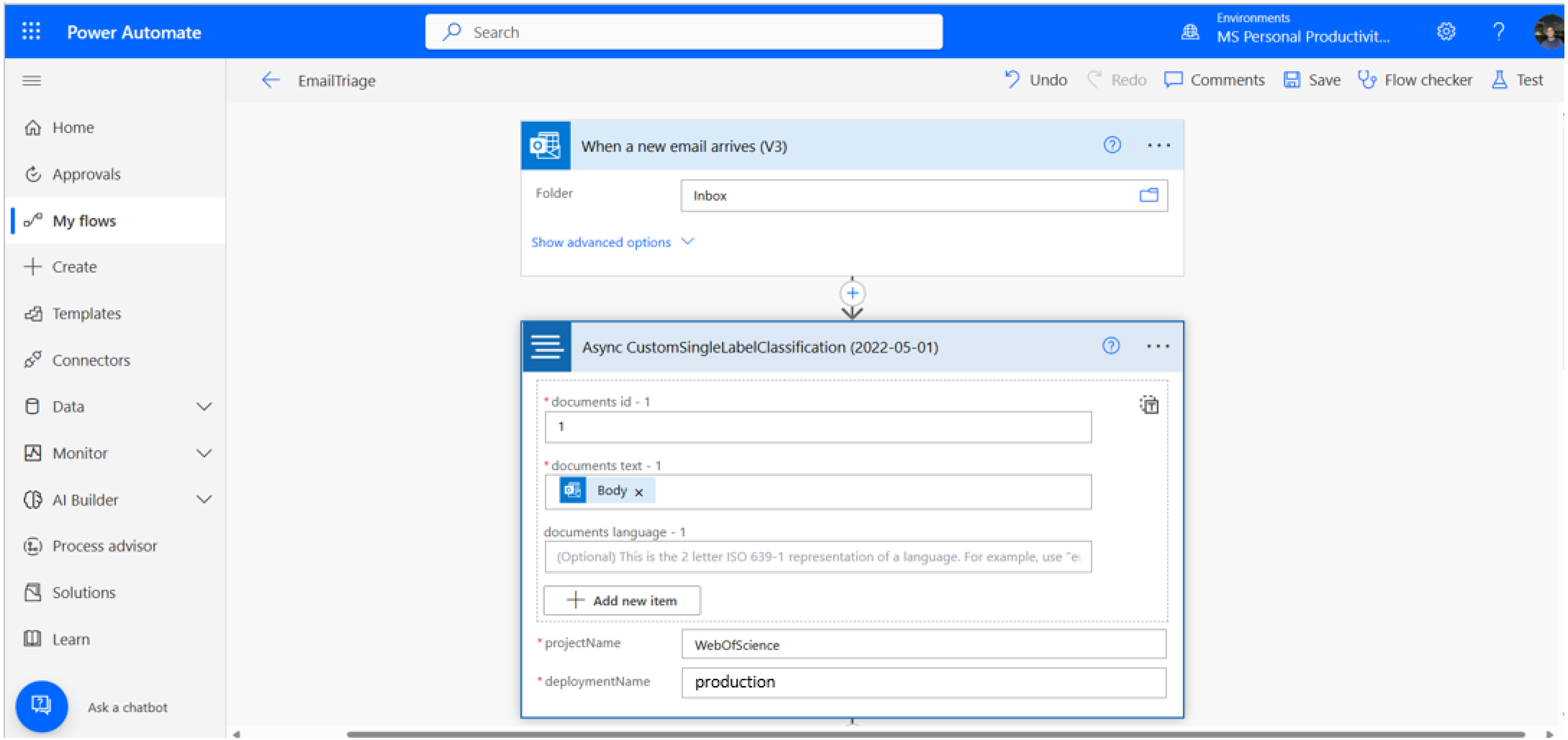
Reference Architecture



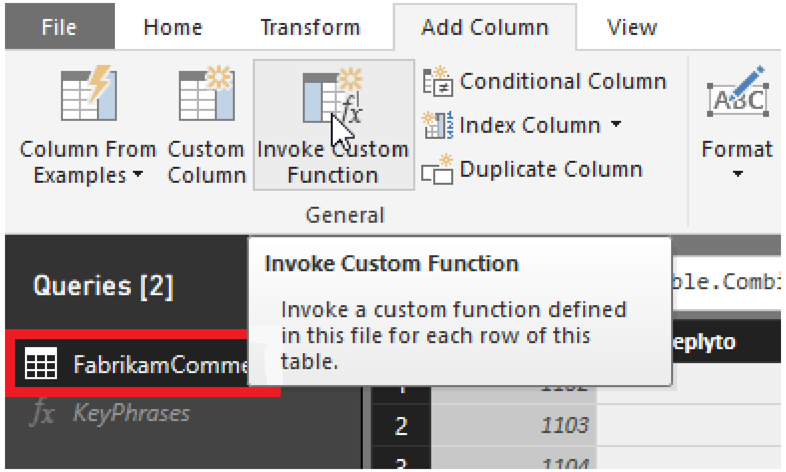
Document Redaction with Azure AI Language PII Native Document Analysis



Triage incoming emails with custom text classification and power automate



Extract key phrases from text stored in Power BI



Microsoft

https://aka.ms/language-accelerators

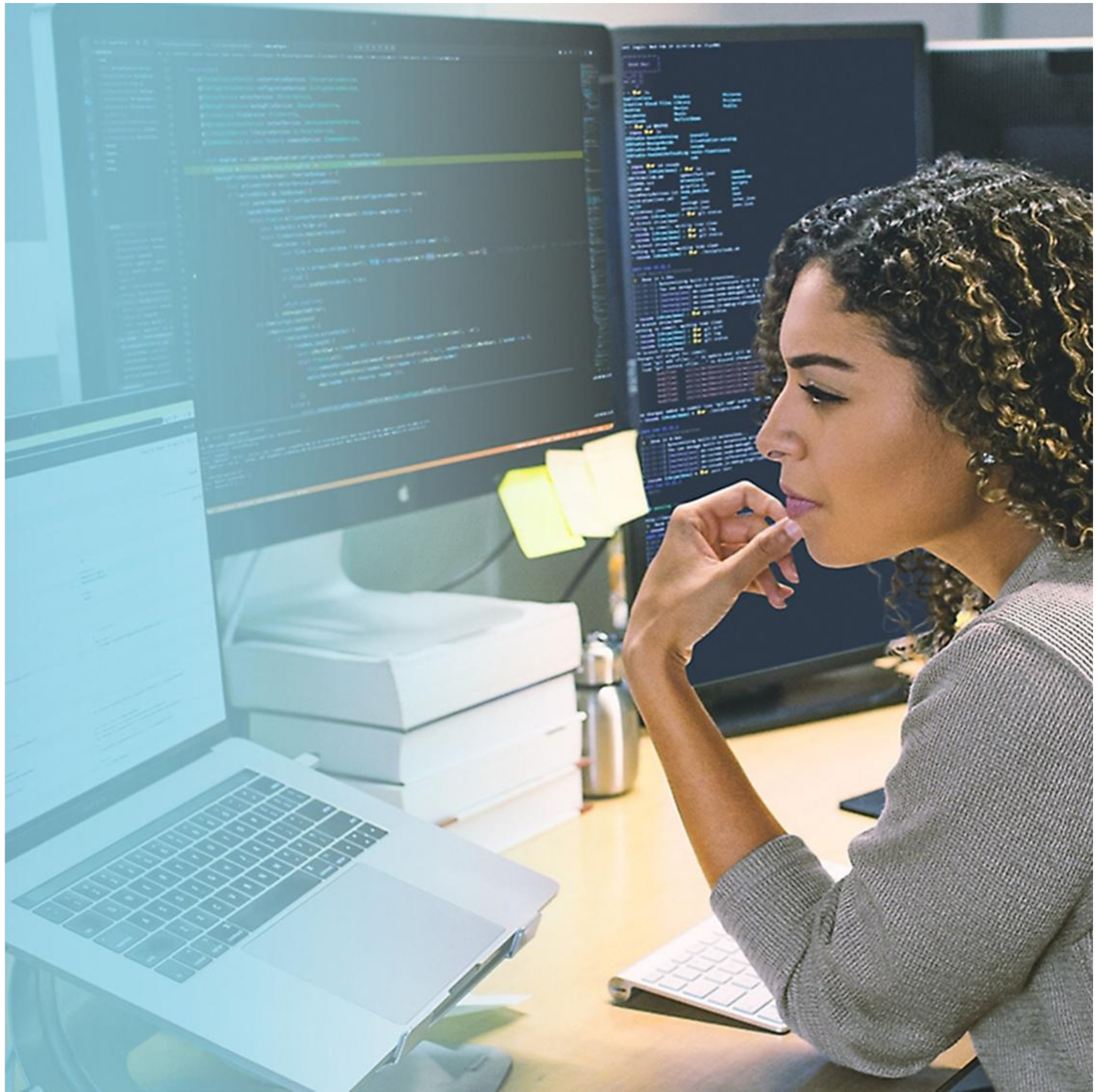
Beirrsdorf

Azure AI Language Azure AI Search Azure AI Services

Beiersdorf

"We're able to free our researchers from interacting with dozens of different interfaces, which means more quality time spent on research and innovation."

Stephan Abend, Head of the Data Science Hub, Beiersdorf



Arthur Little

Azure AI Language   Azure OpenAI Service   Azure AI Search

ARTHUR D LITTLE

"We were surprised that Microsoft Azure offers such a breadth of advanced AI capabilities, like abstract summarization, which are way beyond even the specialist AI providers"

- Jon Nicholls, Global CIO, Arthur D. Little

Datasite

Azure AI Language | Azure AI Translator | Azure AI Services

Datasite®
Where deals are made

“Sellers might have up to 100,000 documents associated with a deal.... AI features enabled by Azure AI Language can potentially compress weeks of work into days for our customers.”

Thomas Fredell, Chief Product Officer, Datasite

Task Agents

Azure AI Foundry

Copilot Studio

Visual Studio

GitHub

Azure AI Foundry SDK

Model Catalog

Foundational models | Open-source models | Task models | Industry models

Azure OpenAI Service

Azure AI Search

Azure AI Agent Service

Azure AI Content Safety

Azure Machine Learning

Evaluations | Customization | Governance | Monitoring

Observability

Agentic AI

A new frontier

What are agents?

AI designed to perform a task

Tasks can vary in level of complexity and capabilities depending on your need

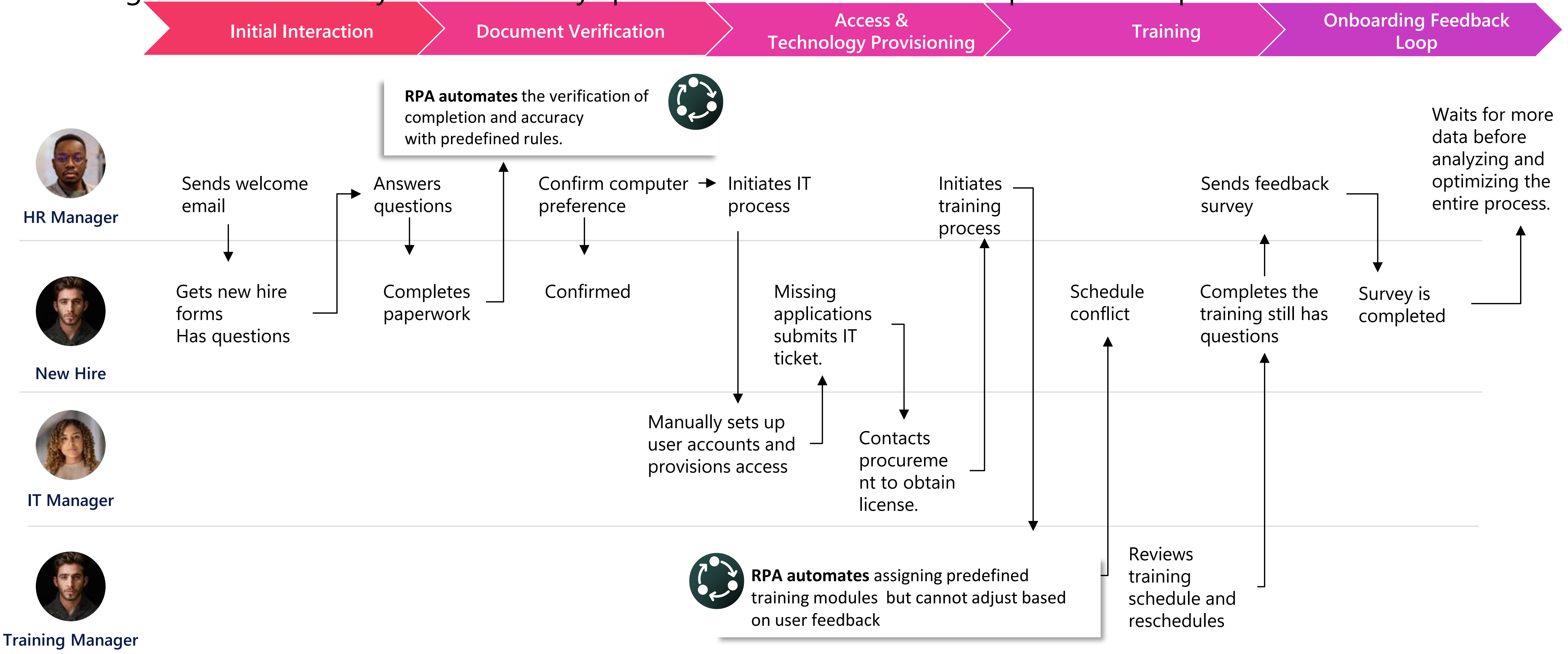
Simple — Advanced

Generation	Retrieval	Action
Generate summaries, images, audio, and more with an AI model and inputs.	**Retrieve information** from grounding data, reason, summarize, and answer user questions	**Take actions** to automate workflows, and replace repetitive tasks for users
Generally available	Generally available	Generally available

Microsoft

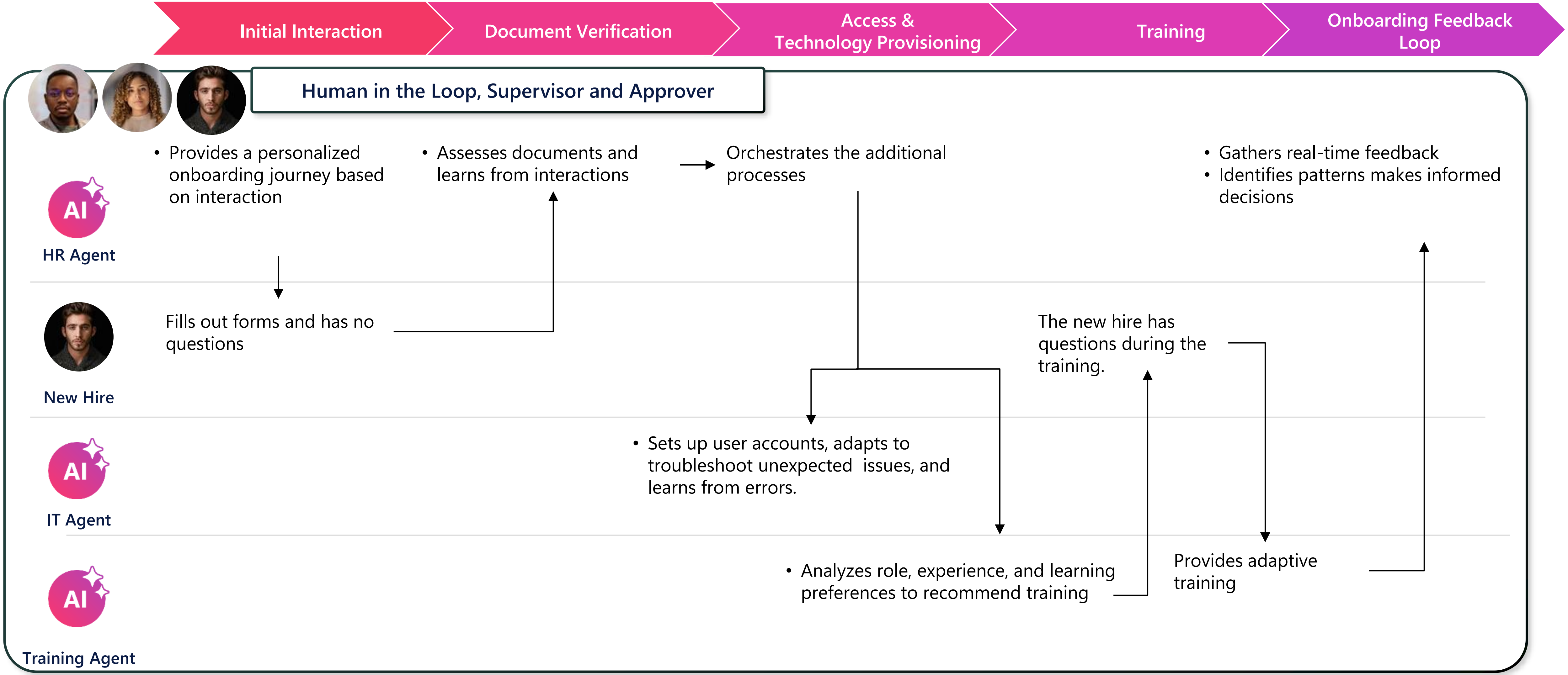
TODAY

Existing solutions can only automate very specific tasks that have clear inputs and outputs

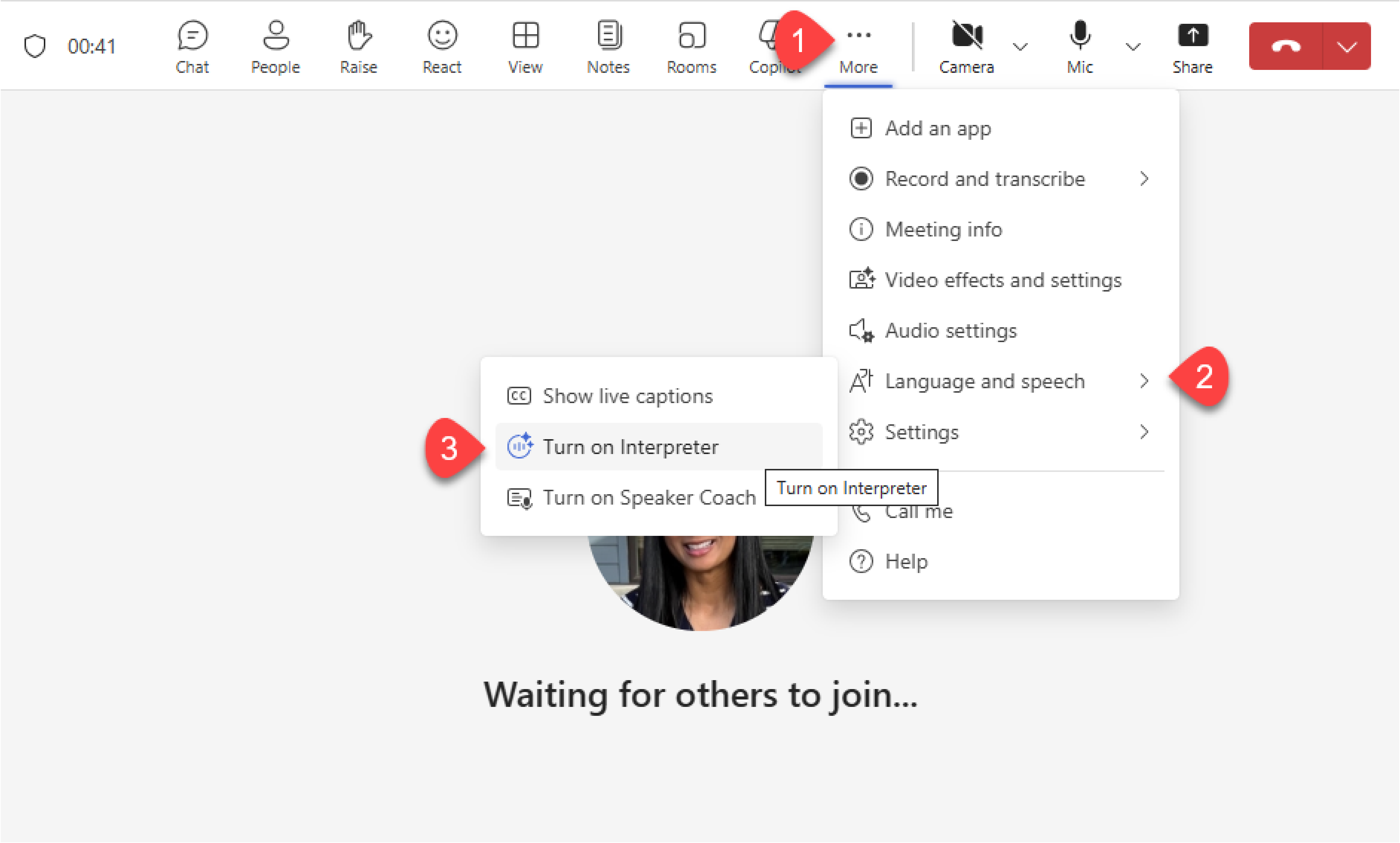


TOMORROW

With AI Agents, these steps can be automated for the first time, while keeping human in the loop.



Interpreter Agent in Teams



Learn More

- Azure AI Language homepage: https://aka.ms/azure-language
- Azure AI Language product documentation: https://aka.ms/language-docs
- Azure AI Language product demo videos: https://aka.ms/language-videos
- Explore Azure AI Language in Azure AI Studio: https://aka.ms/AzureAiLanguage

Self-paced Training

- [Certification: AI Fundamentals]
- [Learning Path: Azure AI Services]
- [Learning Path: Natural Language Processing]
- [Module: Monitor Azure AI services]

Lab Introduction

Lab Objectives

After completing this Learning, you will be able to:

1

Understand....

- ✓ How to call the AI Language service through a low code mechanism (Logic Apps)
- ✓ How to integrate Logic Apps calls with the AI Agents service in Azure AI Foundry